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Population Cycles in Voles and Lemmings: Density Dependence and Phase Dependence in a Stochastic World

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Nils Chr. Stenseth



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I review the regular multiannual population fluctuations in voles and lemmings of northern latitudes. Periodically fluctuating small rodent populations all exhibit a clear two-dimensional density-dependent structure. This implies both a direct and a delayed annual density dependence, and suggests that either a predator-prey type of interaction or a specialised plant-herbivore type of interaction (but not both) *may* be the underlying cause of these multiannual density cycles. Clear-cut experimental testing relating to these propositions is, however, lacking. A two-dimensional annual density-dependent structure is typically non-linear in a way which may be modelled as a threshold type of non-linearity and interpreted as a phase dependence in the density-dependent structure (implying that the density-dependent structure is different in the increase and the decrease phase). Two clear geographic gradients in the annual density-dependent structure are reviewed: the Fennoscandian gradient and the Hokkaidian gradient. The seasonal nature of the annual density-dependent structure is furthermore reviewed: for voles in both Fennoscandia and Hokkaido the direct annual density dependence during the winter (measured per time unit) is concluded to be the strongest. I close with a survey of the main challenges within the field of 'small rodent cycles' – the greatest of which is suggested to be the integration of demography and population dynamics. I recommend to look at other populations for potentially applicable model systems. I conclude that multiannual population cycles seen in voles and lemmings will continue to be a strong source of conceptual and methodological developments within the field of population ecology.

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Although natural historians and laymen have long been aware of the dramatic and fairly regular fluctuations in the abundance of rodents and other terrestrial vertebrates of northern regions, it was Charles S. Elton who in 1924 made the scientific community broadly interested in 'the rodent cycle' (see, e.g., Stenseth and Ims 1993a, Stenseth 1995a). Seventy-five years later – and 50 years after the first issue of *Oikos* – we know much more about this phenomenon than did Elton and his contemporaries. In particular, we have obtained a good understanding of the multiannual *structure* of the vole and lemming cycles. However, we still do not know

why these regular density fluctuations occur – but much seems to suggest that specialised trophic interactions may play a key role in their generation.

In this Anniversary review I will survey what we currently know about these regular population fluctuations in voles and lemmings of the north. Having been involved in this field for the last 25 years, I have chosen to make this a rather personal account of the field. Profound advances have been made over these 25 years: First, we now have many more – and longer – time series on annual (as well as seasonal) abundances in voles and lemmings than we had 25 years ago. This

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advancement is to some extent due to the Nordic (and Eltonian) ecologists' collecting tradition within which loyalty has been paid to annual and biannual sampling protocols in spite of funding agencies not always being supportive. Second, our ability to analyse such time-series data have been much improved thanks to a revived interest from statisticians in the phenomenon of periodic population fluctuations, and as a result of this to the development of new statistical methods (see, e.g., Royama 1992, Tong 1990; see below). Third, the way we ask theoretical questions relating to the phenomenon of population cycles is now more directly linked to time-series data and experimentally derived information than 25 years ago. Earlier, theorists (like myself; see, e.g., Stenseth 1977a) asked questions relating to the dynamic consequences of given sets of assumptions about how individuals interact – an approach nicely falling within the tradition originally developed by Lotka (1925, 1932) and Volterra (1926). Today, theorists (like myself; see, e.g., Stenseth et al. 1996a, 1998a) ask what underlying mechanisms might possibly generate the statistically deduced temporal (and spatial) structure in the multiannual population cycles. Finally, our ability to perform field experiments has become greatly improved over the last 25 years – a development which I have, in admiration, been able to observe only from a distance. Indeed, I am not an empiricist myself, but I greatly value the contribution from the experimentalists in this and other ecological fields. Without their input, we would not have had such a good empirical platform as we do have in ecology today.

Today the most exciting challenges – and the most promising developments – within the study of multiannual small rodent population cycles relate to the integration of demographic modelling on an appropriate (i.e., monthly or seasonal) time scale with the annual or seasonal statistically derived population models. More specifically, we may today realistically aim at developing population models of the Leslie–Usher matrix type (see, e.g., Leslie 1945, 1948, Usher 1969; see also Lefkovich 1965, Tuljapurkar and Caswell 1996) based on statistical analysis of capture-mark-recapture data (commonly collected on a monthly basis), and then ask to what extent we can generate a temporal dynamics corresponding to the statistically deduced structure of vole and lemming annual and seasonal time series (see, e.g., Stenseth et al. 1998b, Yoccoz and Stenseth 1999). To some extent this closes the circle: Elton wrote his pioneering 1924 paper discussing time series on lemming eruptions [to a large extent reported by Collett (1911–1912) (see, e.g., Stenseth 1995a)]. This paper of Elton generated much interest in obtaining more and longer time-series data: Elton obviously saw the value of compiling such long-term data. However, Elton clearly also saw the importance of describing the demographic machinery within a population ecological con-

text: “George” Leslie was brought to the *Bureau of Animal Population* (see, e.g., Crowcroft 1991) so as to do his pioneering work within the field of demographic analysis and population modelling (see below). The challenge facing us today is to re-connect demographic and population dynamics analyses, both of which had their initial developments at Elton's Bureau.

However, before I enter into the subject matter of this review, let me make two digressions – the first one on the many heated debates among those working on small rodent fluctuations, and the second one on the year of 1949 (when the first issue of *Oikos* was published).

Digression I: Are ecologists working on voles and lemmings an argumentative bunch whose debates have little relevance to other ecologists – or, is the enigma of the vole and lemming cycles an important but difficult problem to solve?

Ever since Elton's first papers appeared, the causes of rodent cycles have been fiercely debated among ecologists in scientific journals (see, e.g., Lidicker 1988, 1991, Gaines et al. 1991; see also Ostfeld 1998), at meetings, as well as in referee reports on submitted papers. Much of the discussion among rodent biologists resembles ideological quarrels rather than intellectual debates. Through the peer review process, participants have too often tried to censor their opponents' views.

These heated discussions went on at many meetings including the *Cold Spring Harbor Symposium on Quantitative Biology* in 1957 [contributors included Andrewartha (1957), Birch (1957), Calhoun (1957), Chitty (1957), Cole (1957), Errington (1957), Nicholson (1957) and Pitelka (1957)], as well as in symposia organised as part of the *International Theriological Congresses* (Brno, Czechoslovakia in 1960; Helsinki, Finland in 1982; Edmonton, Canada in 1985; and Rome, Italy in 1989), and symposia organised as part of the *European Mammal Congresses* (Southampton in 1995; and Jyväskylä in 1999), and a series of separate meetings [most recently in Grand Forks, US (1996) and Oslo, Norway (1997) (Batzli 1996)]. An important collection of papers appeared in the 1954 volume of *Journal of Wildlife Management* in which Cole (1954), Errington (1954), Hickey (1954), Hutchinson (1954), Lack (1954), Moran (1954), Ricker (1954), Rowan (1954) and Siivonen (1954) helped focus the questions relating to the study of population cycles.

Over the years, the arguments have focused both on whether the multiannual rodent population cycles are real (Cole 1954) and on their causes; the first issue is settled (they are real); the second is – to some extent – still in contention. Explanations of the population cycles have ranged from social behaviour (e.g., Chitty 1952, 1957, 1960, 1967, 1996, Krebs et al. 1973, Krebs 1978, 1996) to food (e.g., Batzli 1985, Hansson 1971), predators (e.g., Pitelka et al. 1955, Pitelka 1973, Pitelka and Batzli 1993, Hansson and Henttonen 1988, Hanski

et al. 1993, Korpimäki 1993, Korpimäki and Krebs 1996) [or trophic interactions more generally (e.g., Hansson 1987, Oksanen 1990)] and disease (Elton et al. 1931, 1935). A sad – but characteristic – feature of the literature on small rodent cycles is that proponents of particular hypotheses have commonly been better at promoting their views on the basis of general verbal arguments, than at providing detailed analysis of long-term data or critical experimental test-results. Too often, rodent ecologists have not properly deduced either the emergent dynamic consequences of assumed interactions within rodent populations, or the consequences of the interactions between rodents and their environment (including predators and/or food). As a result many of the proposed hypotheses on the rodent cycle available in the literature are flawed (see, e.g., Stenseth 1981, 1986). Altogether it is clear that the view that “ecologists working on voles and lemmings are an argumentative bunch whose debates have little relevance to other ecologists” has some basis.

Nevertheless, the periodic fluctuations in small rodents have interested colleagues far outside the group of scientists working on voles and lemmings. The British ecologist Charles S. Elton (1924, 1942) (see also Elton and Nicholson 1942a, b) was indeed highly instrumental in making the population cycles of northern vertebrates such as lemmings and voles (as well as lynx and hares and many other northern species) a key topic within the field of ecology. As an interesting historical point, it should be noticed that the Australian statistician Patrick Moran got to know about the lynx data when he after the Second World War visited Charles Elton and Dennis Chitty in the *Bureau of Animal Population* at Oxford. That visit turned out to be rather influential (see, e.g., Moran 1953a, b; see also Royama 1992, Ranta et al. 1995a, b, 1997a, b, 1999a).

The general relevance of the dynamics of the vole and the lemming cycle is further documented by the discussion on possible differences in dynamics of small and large mammals, where voles and lemming are the prototype small mammal (e.g., Caughley and Krebs 1983, Krebs 1993). The interest of a rather wide-ranging group of ecologists is further documented by the fact that ecologists like Hutchinson (1975, 1978) and authors of several textbooks in ecology (see Stenseth and Ims 1993a) discuss the rodent cycles – even at length. The reason is simple: before we properly can explain why some populations exhibit cycles and others do not, we cannot honestly say that we understand population dynamics.

CODA: The enigma of the vole and lemming cycle remains one of the largest unsolved – and most important – issues within the field of population ecology. Even though ecologists working on voles and lemmings are often seen as an argumentative bunch of people,

much of the work on the small rodent cycle has been of great general interest and relevance to population ecology as such – and will continue to be so in the years to come.

Digression II: Relevant papers of 1949 – the first year of *Oikos*

Over the years, *Oikos* has been a key journal for communicating advances within the field of the rodent cycle (e.g., Hansson 1967, 1984, 1987, 1999, Henttonen et al. 1977, 1987, Andersson and Jonasson 1986, Ydenberg 1987, Ims and Steen 1990, Oksanen 1990, Steen et al. 1990, Korpimäki and Norrdahl 1991, Korpimäki 1993, Åström et al. 1996, Hambäck and Ekerholm 1997, Oksanen et al. 1999). *Oikos* has also published several symposium volumes on the small rodent cycle [see, e.g., Stenseth (ed.) 1977b, Hansson and Stenseth (eds.) 1988]. Unfortunately the only paper published on the small rodent cycle in the first volume of *Oikos* (Palmgren 1949) is not a particularly outstanding one. However, this paper clearly grew out of an important Nordic and Eltonian tradition within ecology, namely the compilation and analysis of long-term data sets on population fluctuations [two of which are shown in Fig. 2: the rodent community, with strong numerical dominance of the grey-sided vole (*Clethrionomys rufocanus*), in the birch forest at Kilpisjärvi (Finland), and the Norwegian lemming (*Lemmus lemmus*) at Finse (Norway)].

Among the relevant papers published worldwide in 1949 (Allen 1949, Burt 1949, Calhoun 1949, Chitty and Kempson 1949, Clarke 1949, Emlen et al. 1949, Hamilton 1949, Hayne 1949a, b, Leslie and Park 1949, Manville 1949, Moran 1949, Palmgren 1949, Solomon 1949), I will single out six papers which have had profound impact on the study of the ‘vole and lemming cycles’ – directly or indirectly: Solomon (1949), Calhoun (1949), Moran (1949), Burt (1949), Hayne (1949a), and Chitty and Kempson (1949).

Solomon (1949) reviewed the mathematical and empirical approaches to population biology – and quite rightly reminds us about mathematics primarily being a tool helping us think more clearly and precisely about ecological processes, whereas only empirical work can tell what really happens. J. B. S. Haldane (1927) in his *Possible worlds* made the same point, as did John Maynard Smith (1982: “Mathematics without natural history is sterile, but natural history without mathematics is muddled”) and Yrjö Haila and Olli Järvinen (1982: “Sound naturalism is to ecology what legs are to a runner; but anti-theoretical naturalists are, quite naturally, like headless runners”). Solomon (1949) also discussed the issue of density dependence and density independence. His discussion is indeed very balanced; however, had his paper been written today – 50 years later – he would, I suspect, have emphasised more that we need some sort of delayed density dependence in

order for density cycles to arise (an issue he does mention in a footnote commenting on previous work by Varley (1947), a comment which is somewhat unclear, I must add).

Calhoun (1949) was among the first to focus on the importance of understanding the social organisation – and dynamics – of rodent populations in order to disentangle the underlying mechanisms of the rodent density cycle (Christian 1950, Voipio 1950, 1988, Chitty 1960, 1967, 1996, Krebs 1964, Krebs and Myers 1974, Stenseth 1981, 1985a, b, 1986, Stenseth and Fagerström 1986). During the last 25 years this field has been among the most active ones – and much debate has taken place around it (for reviews, see, e.g., Cockburn 1988, Krebs 1996).

The papers by Burt (1949) and Hayne (1949a) on territoriality and the estimation of home range size are among the most cited papers within the field of rodent ecology – and certainly among the most cited papers from 1949. These – and their follow-up papers – have led to an extensive literature on the issue of space use (see, e.g., Stenseth and Lidicker 1992, Andreassen et al. 1993).

The note by Moran (1949) on the lynx cycle signifies the entry of statisticians into the study of population cycles in northern vertebrates. Since then several statisticians, including Moran (1953a, b) himself, have played an important role in improving our understanding of the temporal and spatial structure of the small rodent cycle through statistical modelling of ecological time-series data. In the following section, I provide a short and elementary introduction to time-series modelling of census data and their like. I will focus on annual survey data and log-linear models [much of which will follow closely the nice exposition made by Royama (1992)]. In a subsequent section I will discuss two non-linear time-series models for small rodents.

At a more practical level, the paper by Chitty and Kempson (1949) is a very important one since it described a new trap which came to be called the ‘Longworth trap’ (so named since it was produced by Longworth Scientific Instrument Company; see Chitty 1996) – the Anglo-Saxon analogue of the Scandinavian Uggan trap (Hansson 1967) being extensively used all over Europe. These traps have provided the basis for many live-trapping studies which again have provided a good basis for estimating population abundance (cf. Emlen et al. 1949, Hayne 1949b) and demographic parameters (cf. Leslie and Ranson 1940, Leslie and Park 1949, Leslie and Chitty 1951, Leslie 1952, Leslie et al. 1953) through capture-mark-recapture modelling (cf. Manville 1949; see also, e.g., Yoccoz et al. 1993, 1998, Julliard et al. 1999, Prévot-Julliard et al. 1999).

CODA: Ever since Elton’s pioneering paper of 1924 there has been a growing literature on small rodent density cycles – a growth which has been particularly

pronounced during the 50 years Oikos has existed. This growing literature on small rodent cycles has played – and continues to play – an important role in the development of general ecological theory on population dynamics.

A methodological preamble: autoregressive modelling of small rodent time series – and its interpretation

Annual time-series data (most commonly corresponding to the autumn population) are of the form $N_1, N_2, \dots, N_n$, where each of the N ’s typically is the number of individuals caught per trapping effort (or in some cases estimated population sizes) in the particular year, and n is the number of years for which we have such annual data. Such data are [for statistical reasons; in order to stabilise the variance (see, e.g., Stenseth et al. 1996b)] often log-transformed, possibly after a fixed number (e.g., $\gamma = 1$) is added in order to avoid ‘0-observations’; i.e., $x_1, x_2, \dots, x_n$, where $x_t = \log(N_t + \gamma)$. [One option of avoiding the zero problem, and probably the most natural one from a classical statistical point of view, would be to formulate a generalised linear and additive model with a Poisson or Gamma error term (see, e.g., Bjørnstad et al. 1998a). Currently we (Bølviken, Gløckner and Stenseth unpubl.) are taking another approach also avoiding this problem by incorporating a separate observation model superimposed upon the ecological model (see also Fromentin et al. 1999).] Even though the log-transformation is primarily done for statistical reasons, it also has some very favourable ‘ecological’ features (see, e.g., Williamson 1972), one of which being to account for the commonly observed feature that adding one individual to a sparse population may have a much larger effect than adding one individual to a dense population.

A log-linear autoregressive model of order two [i.e., with two lags, as is typically observed for periodically fluctuating small rodent time series; see, e.g., Stenseth et al. 1996a, b)] is given as:

$$x_t = a_0 + a_1 x_{t-1} + a_2 x_{t-2} + \varepsilon_t, \quad (1)$$

where ε_t is some noise variable incorporating among other things environmental stochasticity (cf. May 1973), and a_1 and a_2 are the first and second autoregressive coefficients, respectively (a_0 is only a location or translation parameter with no dynamic effects); the former of these (a_1) is referred to as the direct annual density dependence whereas the latter (a_2) is referred to as the delayed annual density dependence. Notice, however, that the statistical parameter a_1 , relates to what might be called the ecological direct annual density dependence, b_1 , through the following formula: $a_1 =$

$1 + b_1$. This can be seen by formulating the following population dynamics model:

$$N_t = N_{t-1} \exp(b_0 + b_1 x_{t-1} + b_2 x_{t-2} + \varepsilon_t) \quad (2)$$

and taking logs on both sides and collecting terms, we obtain:

$$x_t = b_0 + (1 + b_1)x_{t-1} + b_2 x_{t-2} + \varepsilon_t. \quad (3)$$

Hence, no direct annual density dependence in an ecological sense (i.e., $b_1 = 0$) will correspond to $a_1 = 1$; stronger ecological direct annual density dependence (i.e., b_1 being increasingly more negative) will then correspond to a_1 being progressively less than one.

The quantities $\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \varepsilon_{t-3}, \dots$ in Eq. (1) are symmetric and independently distributed stochastic variables with constant mean and variance. Moran (1953b) was the first to fit a log-linear autoregressive model as given by Eq. (1) to ecological data (on lynx). He was, however, well aware of the inadequacy of this model. In particular, he pointed out the inhomogeneity of the fitted residuals, violating the assumption of a common and constant variance for the white noise term, ε_t , in the fitted model. It is, nevertheless, an observational fact that a log-linear autoregressive model of order two has helped a lot in interpreting the wealth of time-series data we have access to today – a point I hope to demonstrate throughout this review.

The dynamic properties of this model are well understood (see, e.g., Brochwell and Davis 1991; see also Royama 1992, Bjørnstad et al. 1995) and illustrated in Fig. 1 which represents the parameter space of the statistical model given by Eq. (1): outside the triangle, there is non-stationary dynamics – hence real (and sustained) populations must fall within the triangle. Under the parabola, there will be (proper) multiannual cycles with varying dominating periods (defined as $1/\omega$ where ω is the frequency corresponding to the peak in the estimated spectral density function; see Fig. 3, below) – the length of which will depend on the autoregressive parameters. [Below the parabola the characteristic polynomial is imaginary (and hence the system is intrinsically oscillatory).] In the upper left part of the triangle there will be a two-year oscillation due to strong direct annual density dependence (i.e., $a_1 < 0$ and $b_1 < -1$) – a dynamic pattern which, as far as I am concerned, shall not be considered to correspond to proper multiannual small rodent cycles (they are too short and do not have the flexibility of changing the length of the cycle in response to environmental conditions). In order to facilitate the interpretation of several of the diagrams used in this review, I have included the diagram shown in Fig. 1 as a small insert into several of the other figures in this paper.

It follows that under the assumption of a log-linear model, we need a second-order model (i.e., a model

with a second lag) in order to generate the proper population cycles seen in many lemmings and many other small rodents of the north. If the underlying process is strongly non-linear, we may no longer need a second-order delay in order to obtain sustained cycles (see, e.g., May 1974). All vole and lemming populations for which we have long-term data – and exhibiting multiannual density cycles – may properly be concluded to show a second-order structure with delayed annual density dependence.

Assuming fixed parameters (i.e., fixed a_1 and a_2), the model given by Eq. (1), including noise, gives rise to what may be called ‘sustained cycles’ with relatively fixed period and greatly varying amplitudes [just as is typically observed in small rodent cycles; see, e.g., Stenseth and Ims 1993b]. It should be noticed, though, that such a model – with fixed parameters – frequently gives rise to long time-segments with ‘atypical’ fluctuating dynamics. Having observed this, I remain sceptical about statements in the literature claiming that the cyclic density changes recently have disappeared (see, e.g., Hansson 1999). I am not saying that the cycles have not disappeared in northern Fennoscandia (as has been claimed; see, e.g., Hansson 1999). All I am saying is that we better be careful in making too strong statements before the proper statistical analysis has been done [e.g., as exemplified by Stenseth et al. (1999) working on the common structure in panels of lynx populations].

Fluctuating small rodents are indeed good model organisms for studying the type of oscillatory dynamics shown in Fig. 1 – a fact which may explain why fluctuating small rodents have played, and continue to play, such an important role in general ecology. Part of the reason for voles and lemmings of the north continuing to be good model systems is clearly due to the fairly large amplitude and the fairly fixed period (e.g., Krebs

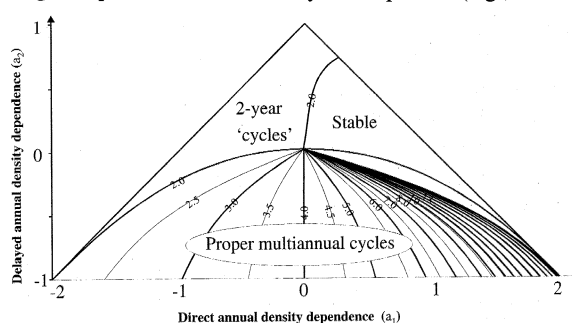


Fig. 1. The dynamics of the statistical autoregressive model given by Eq. (1), where the direct annual density dependence is given by a_1 and the delayed annual density dependence is given by a_2 (see Royama 1992, Bjørnstad et al. 1995). Under the ‘half circle’ – a parabola – proper cycles will arise (the contour curves indicate the periods for various parts of the statistical parameter space); only within the triangle can populations persist (see Royama 1992). [Figure adopted from Bjørnstad et al. (1995).]

1993, Stenseth and Ims 1993b); i.e., the fact that there are very clear signals in the dynamics.

Within this methodological preamble it is worth asking what type of mathematical formulation is best suited for modelling the population dynamics of small fluctuating rodents – differential equation models (like those used by Lotka and Volterra) implicitly assuming continuous time processes, or difference equation models [like the one given by Eq. (2)]. The demographic processes generating the emerging population dynamics are certainly continuous in time – but the data we have are typically discrete in time. Hence, discrete-time models seem easier to link to available time-series data – the patterns of which we are to explain. Such models are, I will argue below, also interpretable within the framework of more mechanistic process-oriented explanations incorporating demographic processes, as well as trophic and social processes.

CODA: Log-linear autoregressive statistical models are helpful tools for deducing the patterns of observed population fluctuations. Such autoregressive statistical models may also serve as a good basis for assessing the merits of various proposed mechanisms supposed to underlie the observed population fluctuations.

Defining the ‘cycle’: the numerical approach or the biological approach – a red herring

Krebs (1996, and earlier) has distinguished between what he calls the *biological* definition of cyclic fluctuations focusing on the demographic and physiological characteristics associated with density changes (if the demographic rates are phase dependent, we have a cyclic population, he claims), and the *mathematical* definition of cyclic fluctuations focusing on the large amplitude and multiannual regularity of population fluctuations. Personally, I do not think there need be much uncertainty as to what a vole or lemming multi-annual cycle is: it is the pronounced – and often high-amplitude – multiannual pattern of fluctuations in population densities (i.e., falling under the parabola in Fig. 1). Indeed, Elton (1924) was very clear at this point (e.g., the title of the paper being “Periodic fluctuations in the numbers of animals”); he says on p. 125 of that paper: “Evidence is given below that the numbers of lemmings fluctuate periodically...”. The common reference to the lemming cycle as a possible example of what mathematicians refer to as a limit cycle (May 1972, Finerty 1980, Gilbert 1984) is a further documentation of the density variation being the key element. In ecology textbooks it is also clear what a cyclic (rodent) population is assumed to be: Krebs (1994: 285) focuses on the “dramatic cyclic oscillations in density”, and

Begon et al. (1996: 589) state explicitly that one has to study the autocorrelations in the time-series data on abundance in order to ascertain whether or not we are dealing with a population cycle. Hence, either implicitly or explicitly, a mathematical (or rather statistical) definition is used for defining ‘population cycles’ – of course (see also Errington 1954, Hickey 1954).

Having concluded that the population cycle is most appropriately defined mathematically, we should not forget that a whole suite of other changes tend to go along with the density fluctuations – many of which tend to change together with the phase of the cycle. Indeed, the profound contribution of the Chitty–Krebs school (cf. Stenseth and Ims 1993b) is their emphasis on the phase-dependent nature of the density cycles: Chitty (1960, 1967, 1996), Krebs and Myers (1974) and Krebs (1978, 1996) and others (e.g., Christian 1950, 1980, Boonstra and Boag 1987, 1992, Boonstra 1994, Boonstra and Hochachka 1997, Tkadlec and Zejda 1998; see also Cockburn 1988) have all along emphasised the phase dependence of the demography – a pattern, if real, which should be expected to be found in the time-series structure as well (a point I will return to below). Given the historical uses of the term ‘population cycle’, I think it is confusing – and potentially misleading – to refer to ‘population cycles’ as something else than the periodic multiannual density fluctuations (but without forgetting that such a density cycle has associated with it a whole suite of demographic changes). Indeed the contrast between the biological definition of cycle and the mathematical (and statistical) definition of cycle is misleading – they rather represent two sides of the same ‘rodent cycle’ issue.

Krebs (1996) refers explicitly to the *s*-index (cf. Stenseth and Framstad 1980, Henttonen et al. 1985) – defined as the standard deviation of the $\log_{10}$ -transformed abundances in the autumn (for instance) – and sometimes called the ‘cyclicality index’ (e.g., Stenseth and Framstad 1980); see also McArdle and Gaston (1995). This index should not be used as the sole criterion for defining whether a population is cyclic or not, but as far as I can see, it has rarely, if ever, been used as such. However, it is a very useful proxy-variable for differentiating these two classes of dynamics: Henttonen et al. (1985) showed that as short as 5–10-year time-series data may be used for this purpose. [Personally, I regret that Framstad and myself (Stenseth and Framstad 1980) proposed to call the *s*-index ‘an index for cyclicality’. It would have been much better if we had called it an index for variability – which in fact it is. This index also has the problem of making us focus on high-amplitude fluctuations: even though many of the well-known lemming and vole cycles exhibit high-amplitude fluctuations, we could easily be mistaken by excluding low-amplitude fluctuations. In sum, Krebs has a point – but it is a rather minor one, I think].

I hope to document in the following that there is no *real* difference between what is referred to as a 'biological definition' and a 'mathematical definition' of the cycle. Indeed, the analysis suggesting that the non-linearity is a form of phase dependence (see Framstad et al. 1997, Stenseth et al. 1998c; see also below), and the effort to try to see whether the same signature of density dependence and phase dependence is found in both the time-series structure and the demographic rates document that very point.

As suggested by Krebs (1996), interesting and important questions about cycles do include: (1) What produces population declines? (2) What stops population growth in peak years? (3) What determines cycle length? (4) What affects the amplitude of cycles? (5) What causes the phase of low numbers? (6) What affects the rate of population increase? and, (7) What determines the seasonal timing of the decline phase? To this list, Hansson and Henttonen (1985, 1988) added an important question – namely what causes the summer decline (and, as part of that, focusing on the delayed responses)? From my perspective as a population ecologist focusing on the long-term temporal dynamics, I would summarise these questions into the following two major questions: What is the dynamic ecological interaction which generates the fairly fixed periodic nature of the vole and lemming fluctuations with highly variable peak/low ratios from one cycle to the other, and what are the underlying mechanisms making one phase (e.g., the increase phase) different from another phase (e.g., the peak or decrease)? More specifically, knowing that both direct and delayed annual density dependence occurs (which I have already alluded to and which I will document more fully further on in this review), we should also ask: (1) On what demographic component does the direct and delayed annual density dependence operate (remembering that these demographic components can be season-, cohort- or age-specific)? And (2) within which component of the demographic machinery does the memory for the delayed annual density dependence reside? (Similar demographic questions should, of course, be asked regarding the phase dependence of the density-dependent structure, the existence of which I document later in this review.) If we can answer these two sets of questions and on that basis develop demographically based models mimicking the observed properties of the population fluctuations seen in the so-called cyclic small rodents, then we may safely conclude that we have found what the mechanisms for the observed dynamics are. In order to achieve this goal, this means that we need to integrate the findings obtained through the mathematical (and statistical) approach (implied by the so-called statistical definition) with those from the biological approach (implied by the so-called biological definition).

CODA: A population exhibiting a multiannual density cycle should be understood as one whose density (measured, e.g., in the autumn or some other fixed time) shows clear multiannual periodic oscillations during the cycle. The net population growth rate may change through the cycle depending on density (i.e., annual density dependence; direct and delayed), phase (i.e., phase dependence) or both (see below). Similar density-dependent and phase-dependent structures are expected to occur in the demographic rates (see below).

Some definitions

In the following, I will use multiannual fluctuations or periodic fluctuations to mean regular fluctuations with fixed periods of more than a year between peaks. This is referring to what in the literature often is called the vole or lemming cycle (or density cycle), which is the core focus of this review. This then is to be differentiated from seasonal fluctuations and irregular fluctuations.

When referring to density dependence (direct or delayed) I will, whenever appropriate, use the term annual density dependence to emphasise that most of this discussion applies to analysis of annual density data. Whenever appropriate, I will use seasonal density dependence and per time density dependence. I believe these specifications are important and are largely neglected in the literature.

Finally, by 'demography' I will refer to the study of reproductive and survival rates, whereas by 'population dynamics' I will refer to changes in density (and composition) of the population over time.

The empirical basis – the long-term data on fluctuations in voles and lemmings

When Charles S. Elton in 1923 first skimmed through the pages of Robert Collett's book *Norges pattedyr* (*Norwegian Mammals*), he immediately realised that the lemming dynamics seen in the Norwegian mountains exhibited some sort of periodic fluctuations, with peaks every three to four years (Charles Elton pers. comm.; see also Stenseth 1995a). He then believed (wrongly, we now know) that the vole fluctuations seen in the surroundings of Oxford were of the same kind. Today we know that the structure of the fluctuations seen in the Norwegian lemmings (*Lemmus lemmus*) and other northerly distributed rodent populations are distinctly different from the fluctuations seen in the bank vole (*Clethrionomys glareolus*) of Wytham Woods outside Oxford [see Fig. 2 (compare, e.g., F and H) and Fig. 3 (compare, e.g., A and D)]. As can be seen from Fig. 3,

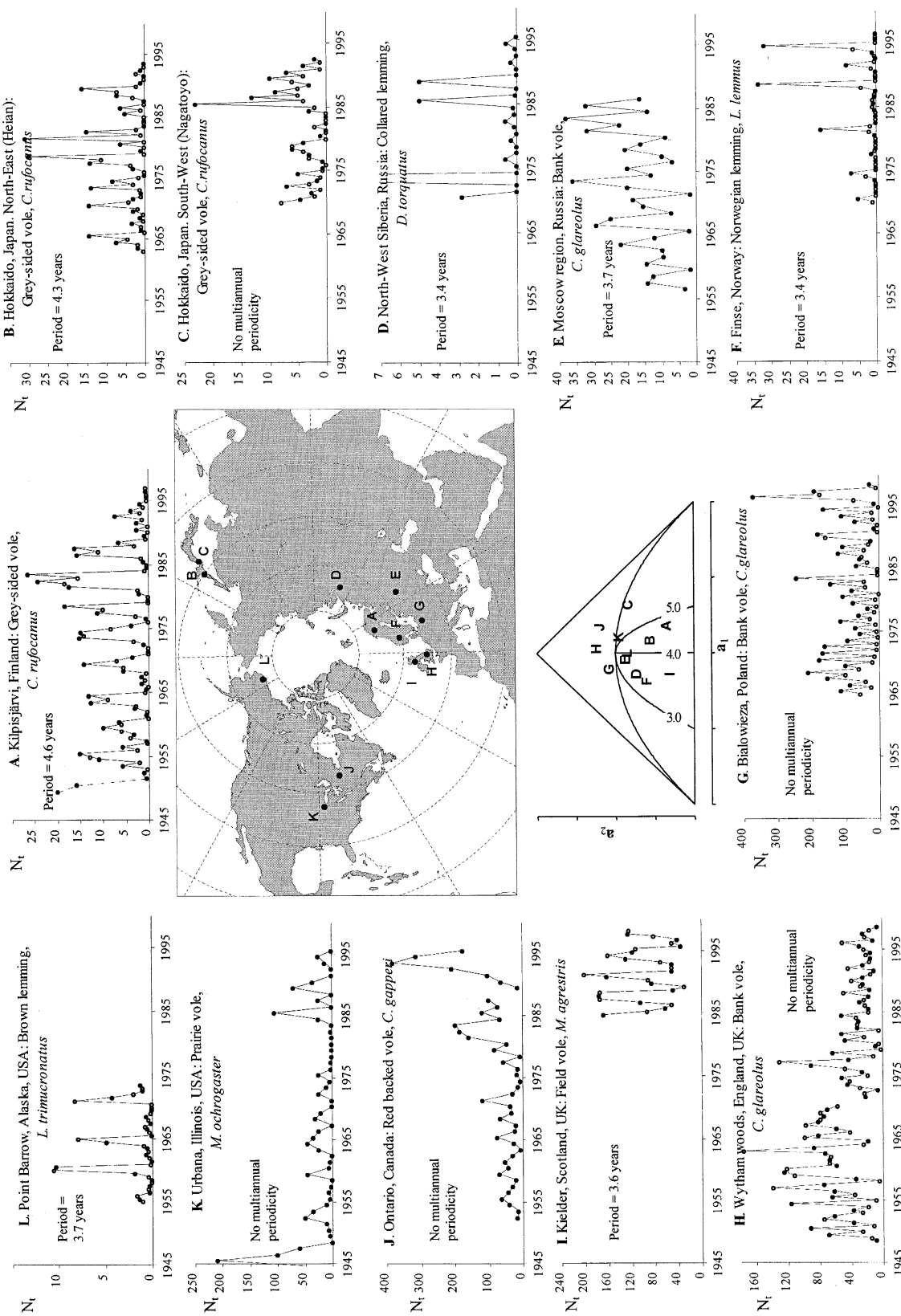


Fig. 2.

Fig. 2. Fluctuations in voles and lemmings: open circles correspond to spring samplings and filled circles correspond to autumn samples. Sites from where each of the time series originates are shown on the map (letters correspond to the letters of the corresponding time-series panel). Below the map is shown the location of the various time series within the autoregressive parameter space (of the type shown in Fig. 1), below the 'half-circle' proper multiannual population cycles is concluded to occur; periods 3, 4 and 5 years are indicated, as is the curve separating two-year cycles and non-oscillatory dynamics. Within each of the panels is given the estimated multiannual period (whenever applicable), being calculated by the method described by Priestley (1981). A: the grey-sided vole (*Clethrionomys rufocanus*) at Kilpisjärvi in Finnish Lapland, 1949–present (Hansen et al. 1999a, Henttonen and Hanski 1999; see also Laine and Henttonen 1983, 1987); B: the grey-sided vole (*C. rufocanus*) in Hokkaido (Heian) representing a multiannually cyclic population, 1962–1993 (Saitoh et al. 1998); C: the grey-sided vole (*C. rufocanus*) in Hokkaido (Nagatoyo) representing a population with only seasonal fluctuations, 1970–1992 (Saitoh et al. 1998); D: the collared lemming (*Dicrostonyx torquatus*) at the Yamal Peninsula, North-West Siberia, 1970–1995 (Balahonov et al. 1997); E: the bank vole (*Clethrionomys glareolus*) west of Moscow, Russia, 1956–1986 (see Smirnov 1970, Bashenina 1981, Ivankina 1987); F: the Norwegian lemming (*Lemmus lemmus*) at Finse in the mountains of southern Norway, 1970–present (Framstad et al. 1993a, b, 1997, Stenseth et al. 1998c); G: the bank vole (*C. glareolus*) in Białowieża Natl. Park, Poland, 1965–1998 (Pucek et al. 1993; Z. Pucek, W. Jedrzejewski and B. Jedrzejewska unpubl.; see also Jedrzejewski and Jedrzejewska 1996); H: the bank vole (*C. glareolus*) from Wytham Woods 1948–present (Southern 1979; and J. Lloyd pers. comm.); I: the field vole (*Microtus agrestis*) from Kielder forest, Scotland, UK, 1984–1998 (Lambin et al. 1999) [This is the classic site previously studied by Middelton (1931) and Elton et al. (1935) from 1931–1939; see Chitty (1996)]; J: the red backed vole (*Clethrionomys gapperi*) from Algonquin Provincial Park, Ontario, Canada, 1952–1995 (Fryxell et al. 1998); K: the prairie vole (*Microtus ochrogaster*) from Urbana, Illinois, USA, 1945–1995 (Getz and Hofmann 1999); and L: the brown lemming (*Lemmus trimucronatus*) close to Point Barrow, Northern Alaska, 1955–1973 (Pitelka and Batzli 1993; F. Pitelka pers. comm.).

there are essentially two kinds of fluctuations among voles and lemmings – those with *only* seasonal density fluctuations (such as the Wytham Woods bank voles; see, e.g., Fig. 3D) and those with *both* multiannual and seasonal density fluctuations (such as the Kilpisjärvi populations of the grey-sided vole (*C. rufocanus*; see, e.g., Fig. 3A); in this review I focus on the multiannual density fluctuations.

The same species may exhibit different patterns of fluctuations (e.g., only seasonal fluctuations vs both seasonal and multiannual fluctuations) in different parts of its range of distribution: compare, e.g., the multiannually fluctuating grey-sided vole at Kilpisjärvi (Fig. 3A) and in the north-eastern part of Hokkaido (Fig. 3B) and the only seasonally fluctuating grey-sided vole in the south-eastern part of Hokkaido (Fig. 3E; here a long term trend is found). It was Lennart Hansson (1971) who first made the point that the same species may exhibit different patterns of fluctuations over its geographic range¹. Today we know, for instance, that in Fennoscandia, vole populations north of approximately 60°N exhibit multiannual density fluctuations, whereas populations further south only exhibit seasonal fluctuations (see, e.g., Hansson and Henttonen 1988; see also Figs 2, 3 and 6). In Hokkaido a similar geographic pattern is observed (see, e.g., Bjørnstad et al. 1996, Stenseth et al. 1996b, Saitoh et al. 1998; see also Figs 2, 3 and 7). Even though the Fennoscandian gradient originally was documented separately for two species [the field vole (*Microtus agrestis*) and the bank vole (*C. glareolus*)] the Fennoscandian gradient is today generally discussed with reference to several species as a group (for which many things may vary). The Hokkaidian gradient, on the other hand, refers to one particular species (the grey-sided vole). The range of habitats is furthermore much greater in Fennoscandia [ranging from alpine meadows at Finse to agricultural land in Alajoki (see Fig. 6)] as well as geographically more extensive than in Hokkaido (see Hansson and Henttonen 1998). It should furthermore be noticed that we are hard pressed to find many examples of non-cyclic small rodent populations in the Arctic: the only example I know is the one studied by Krebs and colleagues (see, e.g., Krebs et al. 1995, Reid et al. 1995; see also Krebs 1996) – and that population may be considered to have exceptionally low density (possibly due to

¹ It seems that Fuller (1969) appreciated that a given species of the typical cyclic kind not necessarily was cyclic all across its range of distribution. He said for instance: "Such a finding would reinforce the author's hypothesis (Fuller 1967) that cycles are fundamentally a phenomenon of regions with long snowy winter, and that the explanation should be sought in a study of the ecology of fluctuating species during the snowy period of the year". However, there is no question that Hansson (1971) was the first to fully appreciate the importance of this – and that Hansson and Henttonen (1985) made this insight generally appreciated by a larger audience.

Multiannual and seasonal fluctuations

Only seasonal fluctuations

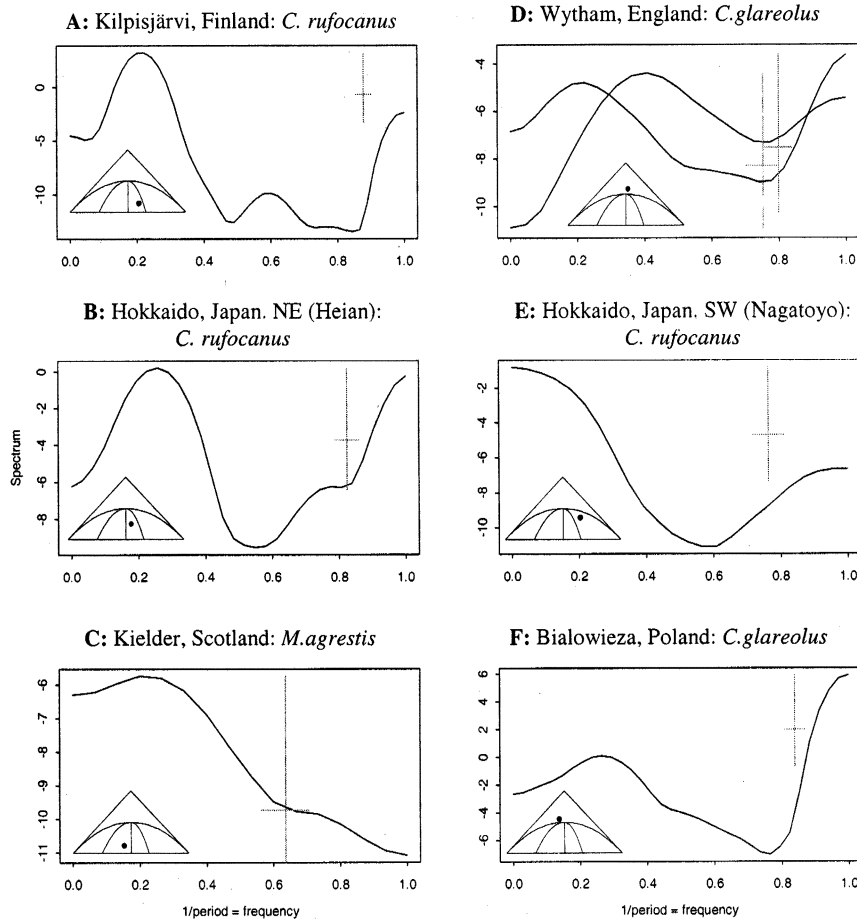


Fig. 3. Smoothed periodograms (Priestley 1981) for some small rodent populations (all of which are shown in Fig. 2), found using *spectrum* in S-plus with $\text{span} = c(5,6)$. For comparison I have inserted within each periodogram the triangular diagram from Fig. 1 with the autoregressive parameters inserted for each of the series; the similarity in pattern deduced from the periodograms and the autoregressive parameter space diagram is, quite naturally, very similar and consistent. Notice, however, that conclusions regarding both seasonal and multiannual periodicity vs only seasonal periodicity is based on the periodograms. The vertical line in the periodogram for a given frequency gives a 95% confidence interval for the indicated frequency. A periodogram with a peak towards the right (or no peak) signifies a seasonal pattern whereas a peak towards the left signifies a multiannual pattern. A: the grey-sided vole (*Clethrionomys rufocanus*) at Kilpisjärvi in Finnish Lapland, northern Finland, 1949–present (see Hansen et al. 1999a, Henttonen and Hanski 1999); B: the grey-sided vole (*C. rufocanus*) in Hokkaido (Heian) representing a multiannually cyclic population, 1962–1993 (cf. Saitoh et al. 1998); C: the field vole (*Microtus agrestis*) from Kielder forest, Scotland, UK, 1984–1998 (Lambin et al. 1999) (Notice that this series does not seem to have any detectable seasonal dynamic pattern.); D: the bank vole (*Clethrionomys glareolus*) from Wytham Woods, 1948–present (Southern 1979, Judith Lloyd pers. comm.) (this series is split into two since there was a break in the monitoring programme in the early 1970's; notice, however, that both segments show essentially the same pattern); E: the grey-sided vole (*C. rufocanus*) in Hokkaido (Nagatoyo) representing a population with only seasonal fluctuations, 1970–1992 (cf. Saitoh et al. 1998) (the peak of the periodogram towards the very left suggests a long-term trend); F: the bank vole (*C. glareolus*) in Bialowieza Natl. Park, Poland, 1965–1998 (Pucek et al. 1993, Pucek, Z., Jedrzejewski, W. and Jedrzejewska, B. unpubl.).

fairly constant and high predation pressure). However, we should not be surprised to find additional stable populations at isolated places (e.g., islands) in the Arctic.

Figure 2 summarises some of time series available on voles and lemmings; as can be seen, we do now have access to several fairly long time series on voles and

lemmings². We do in particular have a good coverage for Fennoscandia (see Fig. 6) and Hokkaido (northern Japan; see Fig. 7); especially valuable is the set of data from Hokkaido which includes 225 annual autumn time series on the same species, the grey-sided vole (*C. rufocanus*) covering about 30 years – and in addition almost as much data for spring and summer (see Kaneko et al. 1998). In Fig. 2 only few data sets from Siberia are shown – but much more exists from that part of the world. However, until recently Russia has been quite closed – partly for linguistic reasons and partly for political reasons. This state of affairs is now changing and much collaboration between Western and Russian scientists is currently underway. For North America there is, however, a real shortage of long-term time-series data. Some exist (see Fig. 2J, K, L), but nothing comparable to what is available for Fennoscandia and Hokkaido – and Russia. This reflects, I am sure, the fact that North American ecologists working on small rodents have focused on more detailed capture-mark-recapture (CMR) studies (providing information on demographic changes) whereas Fennoscandian, Japanese and Russian scientists have more been emphasising the accumulations of long-term monitoring data; it should be noticed though that Fennoscandian and Japanese mammalogists also have been obtaining large amounts of capture-mark-recapture data (see, e.g., Nakata 1984a, b, Henttonen and Hanski 1999) – much of which is currently being analysed (see, e.g., Yoccoz et al. 1998, Prévot-Julliard et al. 1999). As I will further develop later in this review, one of the challenges within the study of small rodent cycles is to integrate the insights obtained from the survey type of studies providing information on the long-term population dynamics and the more intensive CMR studies providing information on the demographic patterns and processes.

Based on the long-term population data the following pattern emerges (see Fig. 2; see also Stenseth 1985a, Warkowska-Dratnal and Stenseth 1985, Stenseth and Ims 1993b): the period of the cycle tends to be rather fixed at any given site, whereas the amplitude tends to vary quite a bit from one cycle to the next. As can be seen from Fig. 2 there is a tendency for multiannual cycles to occur towards the north – particularly so for the high-amplitude fluctuations; for instance, as pointed out above, in Fennoscandia the multiannual population fluctuations are primarily found north of approximately 60°N (see also Fig. 6). However, somewhat similar multiannual density cycles are also found further south

[see, e.g., the field vole (*M. agrestis*) in Scotland (Fig. 2I)]; however, the low densities in these populations seem generally not to be as low as are those seen in the Fennoscandian populations. Even though mainland European vole populations south of Scandinavia often only exhibit seasonal fluctuations (see, e.g., Fig. 2G) we should notice that clear-cut multiannual density fluctuations also are found further south in Europe, also at lower altitudes (see, e.g., Frank 1957, Jobsen 1988, Mackin-Rogalska and Nabaglo 1990, Salamolard et al. 1999). Regarding the situation further south in Europe, with respect to multiannual cycles or not, we need to be very open-minded. We do not have as many time series from continental Europe as we have from Fennoscandia. My own feeling about the continental European situation is that it is likely to be rather complicated with respect to the geographic pattern in the dynamic structure of small rodents; this being due to the more complicated geography of continental Europe. Furthermore, although the Scandinavian mountain range makes the Fennoscandian situation simpler than in continental Europe, we should not be surprised to see a much more complicated geographic pattern as more data becomes available than currently is reported.

I need to add that over the last 10–15 years there are indications suggesting that the typical multiannual periodic fluctuations have been changed – or disappeared – in northern Fennoscandia (e.g., Henttonen et al. 1987, Hansson 1999, Henttonen 1999). I personally am not convinced about the validity of this claim. First, the statistical documentation supporting such a statement has not yet been provided. Second, it is well-known (see above) that an autoregressive model (including noise) may exhibit ‘deviating’ population dynamics patterns for substantial lengths of time. However, what makes me inclined to accept the claimed disappearance of the typical multiannual cycles of the north is that the changes seem to have occurred in several populations (Hansson 1999). In such a situation we should see a common structure (sensu Stenseth et al. 1999) before and after the claimed change in dynamics – whereas there should be no overall common structure when comparing the before and after the supposed change. Supporting the claimed change in dynamics is also the fact that mechanisms explaining this change have been proposed: Hansson (1999) has, for instance, suggested that the change in dynamics is due to changing agricultural practices and changing landscape structure. Much more remains to be done on this issue – both empirically and theoretically.

CODA: Vole and lemming populations tend – primarily at northern latitudes – to exhibit both seasonal and multiannual fluctuations in density, whereas more southern populations tend to exhibit only seasonal fluctuations although exceptions exist. Only rarely do more northern populations (e.g., the one studied by Krebs et al. 1995) exhibit only seasonal fluctuations.

² One cautionary note should be added with reference to the time-series figures: Since we only have discrete data we are really not allowed to connect the points with lines as I have done. Nevertheless I do so to facilitate us appreciating the temporal structure of the time series. For models, provided they are continuous, we are of course allowed to link the points.

Cycles – reality, numerical artefacts or chaos?

Cole (1954) argued that the claimed ‘population cycle’ – seen, for instance, in small rodents – was nothing but a result of the way we analysed our data – and more specifically due to us averaging observed abundances. Essentially, Cole’s (1954) point is what statisticians are referring to as the Slutsky–Yule effect (see Yule 1926, Moran 1950, Chatfield 1989), namely the introduction of autocorrelation into a time series by some smoothing procedure. If, for example, u_t is a white noise sequence of independent variables with zero mean and fixed variance, then the series v_t obtained as a result of applying a moving average of some order will be an autocorrelated series. The same is true if u_t is operated on by any linear filter (see, e.g., Chatfield 1989).

Even though Cole’s point is well taken, there should be no doubt that the periodic nature of the multiannual population fluctuations is real; that has now been unambiguously demonstrated for several populations of several species (Figs 2, 3; see also Bjørnstad et al. 1995, 1996, 1998b). In Fennoscandia, the period is (across all species) varying from around three years in the southern part to about five years in the northern part (e.g., Henttonen et al. 1985, Hanski et al. 1991, Bjørnstad et al. 1995); a similar pattern is observed in Hokkaido (see, e.g., Bjørnstad et al. 1996, 1998b, Stenseth et al. 1996b). It is also a common observation that the period seems to be more fixed at any given site than the amplitude (see, e.g., Stenseth and Ims 1993b; see also above) – probably due to the profound effect of environmental stochasticity.

Traditionally, the vole and lemming multiannual density cycle has been seen as an example of what mathematicians commonly refer to as ‘limit cycles’ – a closed trajectory in a multidimensional state space (e.g., May 1972, Stenseth 1977a, 1985a). Models showing that such limit cycle behaviour indeed may be expected, for instance for predator-prey type of interactions, abound in the literature (e.g., Hanski et al. 1991, 1993; see also Gilbert 1984). However, it was first demonstrated to be the case for lemmings by Framstad et al. (1997).

In order to assess the temporal nature of the population fluctuations, long-term monitoring data are essential. After the pioneering work of Elton (1942) (and other early workers; e.g., Collett 1911–1912; see, e.g., Stenseth 1995a) much valuable long-term time-series data have accumulated, particularly – but not exclusively – from Fennoscandia (e.g., Siivonen 1948, 1952, 1954, 1957, Hagen 1952, Wildhagen 1952, Koskimies 1955, Kalela 1957, 1962, Myrberget 1973, Pitelka 1973, Hörnfeldt 1978, 1994, Southern 1979, Angelstam et al. 1984, 1985, Henttonen et al. 1985, Hörnfeldt et al. 1986, Linden 1988, Oksanen and Oksanen 1992, Pitelka

and Batzli 1993, Pucek et al. 1993, Lambin et al. 1999, Oksanen et al. 1999); see Fig. 2. These data represent gold mines for those interested in understanding the structure of the seasonal and multiannual fluctuations in lemmings and voles of the north.

After May (1974) demonstrated that even very simple ecological models may have very complicated population dynamics, there has been much interest among ecologists in documenting that real ecological systems – including northern small rodents populations – show chaotic behaviour (e.g., Turchin and Taylor 1992, Turchin 1993, 1995a, Ellner and Turchin 1995; see also May 1986). It is not at all clear whether periodically fluctuating small rodent populations of the north are chaotic (see, e.g., Falck et al. 1995a, b), but it should be remembered that the use of essentially deterministic models (e.g., Turchin and Ellner 1999) to what is – for one reason or another – essentially a stochastic system may, as Sugihara (1994) puts it, be ‘fundamentally flawed’ as a means of detecting chaos. Indeed, it should never be forgotten that ecological systems exist in a stochastic world – and that the stochasticity may have important effects on the dynamics of the system (see, e.g., Nisbet and Gurney 1982, Falck et al. 1995b, Higgins et al. 1997, Gurney and Nisbet 1998, Stenseth et al. 1998a, c, Bjørnstad et al. 1999c). Many of those searching for chaos (e.g., Turchin 1993, Turchin and Ellner 1999) seem to miss this point entirely since their framework is a deterministic one.

Essentially, the discussion relating to chaos is about two issues – the mathematical and the statistical aspects. Chaos is unambiguously defined for deterministic systems with reference to the system’s sensitivity to initial conditions (see, e.g., Tong 1990, 1995). It is, however, not clear how to define chaos in a stochastic system (see, e.g., Tong 1995; see also Falck et al. 1995b). Regarding the statistical issue, it is not clear how much – if anything – relating to chaos can be inferred from empirical observations (which always include some observational error in addition to system stochasticity; see Paladi et al. 1995, Tong 1995, Vulpiani 1995, Loreto et al. 1996). We also need to remember that any conclusion relating to whether a system is chaotic or not is highly dependent upon the model formulation (i.e., the skeleton model) being assumed.

Much methodological work remains to be done before we can address questions regarding chaos in small rodent dynamics (see, e.g., Bartlett 1970, 1990). It seems to me to be extremely premature to say – as Turchin (1993) does (see also Turchin and Ellner (1999) – that small rodents are chaotic or on the ‘edge’ of exhibiting chaotic dynamics.

The issue of chaos in ecological systems also brings in another important conceptual issue: Assuming for the sake of argument that periodically fluctuating populations are chaotic (by some definition or another), it is not at all clear what new biology we would have learnt from such a conclusion: I tend to think, very little –

particularly since we primarily try to understand the very strong periodic component of the fluctuations (see Figs 2, 3; see also Fig. 5 and the corresponding discussion). Above all, we need to understand the underlying mechanisms of the population fluctuations – and in this light it is unclear whether the study of chaos in models will help.

CODA: Northern populations of voles and lemmings show distinct multiannual fluctuations (periods ranging from about three years to about five years). It may not be very helpful to observe that such rodent populations exhibit chaotic dynamics, not least since we never should forget that these systems always have a stochastic component (whose effects we better not forget to ask questions about) as well as a strong deterministic component – essentially resulting in either a combined seasonal and multiannual, or a purely seasonal dynamics. The entire issue of chaos is – in my view – side-tracking and confusing the effort to try to understand the ‘small rodent cycle’ much more than helping us to do so. In sum, what is to be meant by ‘chaos’ in small rodent dynamics is not at all clear – and even if it were, it is not at all clear whether it would help us understand the underlying ecological processes better if we knew that some populations (or all, for that matter) exhibit chaotic dynamics. Indeed, much of the discussions on chaos in the ecological literature (at least) is rather chaotic – it is another red herring.

Direct annual density dependence, delayed annual density dependence and the resulting population dynamics

The discussion over density dependence (i.e., the biotic school) vs density independence (i.e., the abiotic school) has played an important role in population ecology (Thompson 1929, Uvarov 1931, Nicholson 1933, 1957, Andrewartha and Birch 1954, Andrewartha 1957, Itô 1972, 1978, Egerton 1973, Bulmer 1975, den Boer and Reddingius 1996). In the language of this review, this is essentially the debate over the importance of deterministic factors (i.e., biotic factors such as intra- and interspecific interaction) versus stochastic factors (being to a large extent extrinsic abiotic factors such as weather). The winners – in some respect – were the biotic school with their focus on regulation via density-dependent processes, for which the main agents were predators, parasites, diseases, and food shortage (see, e.g., Krebs 1995, Turchin 1995b). But we must never forget that density-independent (and in particular stochastic) processes may influence the system profoundly.

Studies on small rodents have made a significant contribution to the literature on density dependence (see,

e.g., Turchin 1993, Saucy 1994, Stenseth et al. 1996a, b, Saitoh et al. 1997, 1998, 1999). Below, I review some of the studies on small rodents with the objective of addressing what these studies have taught us relative to the dynamic process (with both a deterministic and a stochastic component) generating the regular fluctuations seen in small rodents. It should be emphasised that we are not primarily looking for a test (like most have done), but rather to estimate parameters to be interpreted biologically within a particular mechanistic framework. Time-series analysis (see, e.g., Fig. 1) is an essential tool here. It should be emphasised, though, that we are not primarily interested in the time-series models as such, but we are rather interested in understanding to what extent such time-series models can help us understand the underlying mechanisms of the population dynamics.

The order, d , of the process signifies the number of lags included in the time-series model, expressed in delay co-ordinates: $x_t = f(x_{t-1}, x_{t-2}, \dots, x_{t-d}, \varepsilon_t)$ where x_t is some measure of the population abundance (typically the log-transformed abundance; see above; see, e.g., Royama 1992; see also Stenseth et al. 1996a, b), ε_t is a time-independent white-noise term (i.e., a sequence of independent random variables with mean zero and fixed variance; see, e.g., Royama 1992, Stenseth et al. 1996b), and f is some function – linear, or non-linear, additive or non-additive (see, e.g., Stenseth et al. 1997). For a great number of periodically fluctuating rodent populations, d has been estimated to be 2 (e.g., Turchin 1993, Bjørnstad et al. 1995, Stenseth et al. 1996a, b). Even though the additivity assumption has not been properly tested – due to lack of sufficiently powerful statistical tests – additive models are typically assumed. That is,

$$x_t = f_1(x_{t-1}) + f_2(x_{t-2}) + \varepsilon_t. \quad (4)$$

It is an observational fact that the f -function is quite well (but see below: Fig. 4) approximated, through some Taylor approximation, by a linear function (Stenseth et al. 1996a, b, Saitoh et al. 1997, 1999). That is, Eq. (1) is a good approximation to Eq. (4).

In a totally deterministic world a linear model, like the one given by Eq. (1), cannot lead to sustained periodic density oscillations. That is, within the triangle in Fig. 1 (where populations necessarily must fall), only dampened oscillations will result in a deterministic setting; for a given set of parameter values, there will, however, be a fixed period length under the parabola. But, in the presence of stochasticity, fluctuations of the kind observed in small rodent populations will result [see Stenseth et al. (1996b) who used the residual to access the level of stochasticity]. It should be pointed out though that much theoretical work remains to be done on the issue of how deterministic and stochastic components of natural systems interact – an issue which was recently discussed by Stenseth et al. (1998b, c) and which I will return to below.

For the linear approximation, we may interpret the statistical model $x_t = a_1 x_{t-1} + a_2 x_{t-2} + \varepsilon_t$ corresponding to a trophic interaction model (cf. Stenseth et al. 1996a). That is, let N_t and P_t be the abundance of the vole and its specialised predator (or the food-plant the vole is specialising on). One possible model – which indeed is a good approximation to a class of very general models – for describing the interaction between the two populations is (for the predation variant), in its deterministic skeleton (or ecological model) given as:

$$\begin{aligned} N_t &= N_{t-1} \exp(\beta_{10} - \beta_{11} x_{t-1} - \beta_{12} y_{t-1}) \\ P_t &= P_{t-1} \exp(\beta_{20} - \beta_{22} y_{t-1} + \beta_{21} x_{t-1}), \end{aligned} \quad (5)$$

where $x_t = \ln(N_t)$ – as before – and $y_t = \ln(P_t)$, and the parameters β_{ij} are positive parameters determining the effects of the biological interactions determining the population growth rates of the two species. Taking logs on both sides in Eq. (5) and collecting terms, we obtain (the following skeleton of a bivariate statistical model):

$$\begin{aligned} x_t &= \alpha_{10} + \alpha_{11} x_{t-1} + \alpha_{12} y_{t-1} \\ y_t &= \alpha_{20} + \alpha_{21} x_{t-1} + \alpha_{22} y_{t-1}, \end{aligned} \quad (6)$$

which is a log-linear bivariate model; $\alpha_{10} = \beta_{10}$, $\alpha_{20} = \beta_{20}$, $\alpha_{11} = (1 - \beta_{11})$, $\alpha_{22} = (1 - \beta_{22})$, $\alpha_{21} = \beta_{21}$, $\alpha_{12} = -\beta_{12}$.

As is commonly the case, we typically only have data on one of the assumed interacting species. We may, for instance, only have data on the rodent (i.e., on N_t). We may in such cases rewrite Eqs (5) and (6) as an equation in delay coordinates of the species we do have data on (i.e., in our assumed case, the rodent, i.e., x_t)³. That is, we obtain the skeleton of the statistical model expressed in delay coordinates,

$$\begin{aligned} x_t &= (\alpha_{10} + \alpha_{20}\alpha_{12} - \alpha_{10}\alpha_{22}) + (\alpha_{11} + \alpha_{22})x_{t-1} \\ &\quad + (\alpha_{12}\alpha_{21} - \alpha_{11}\alpha_{22})x_{t-2}. \end{aligned} \quad (7)$$

This is a model having the same form as Eq. (1) – i.e., an order-two process, but now with the ecological parameters (α 's) explicitly incorporated in the statistical model-structure.

Even though an order-two process *may* correspond to either a specialised plant-herbivore interaction or a specialist predator-prey interaction, an order-two process cannot correspond to a system where *both* food and predators interact with the voles in a dynamic way exhibiting closed feed-back loops. If *both* food and predators interact with the small rodents in a dynamic manner (e.g., a three-trophic system), then an order-three

process will, in the linear case, result (see, e.g., Stenseth 1995b, Stenseth et al. 1997, 1998d). It might be claimed that specialised predator interactions with voles and lemmings might be important during some period of time, whereas interactions between voles (and lemmings) and their specialised food items are important during other periods (e.g., Keith 1963, 1990). If this view is right, we should indeed expect to observe an order-three process; hence, such a view is incompatible with what we do observe. It should finally be added that an order-two model not necessarily corresponds to a specialist trophic interactive system. Experience shows, however, that it is difficult to see how other processes can generate sufficiently long delays in the interaction so as to give rise to an order-two autoregressive process. Nevertheless, other possible mechanisms should not be overlooked. Personally, I think, however, pathogen-host interactions as a possible mechanism are greatly overlooked (see, e.g., Stenseth 1985a). This was originally suggested as a mechanism for explaining the cycles by Elton et al. (1931, 1935), but never really taken seriously – possibly because we did not have the appropriate technology at the time [a technology we now have; see, e.g., Feore et al. 1997, Begon et al. 1998, 1999]. I am much more sceptical about the age-structuring of the populations being such a possible mechanism (see, e.g., Matthiopoulos et al. 1998, Jorde et al. 1999; but see Wikan and Mjølhus 1995, Oli and Dobson 1999)⁴ – this because of the short life span of small rodents.

Tentatively we may conclude that an order-two process seems to suggest that voles and lemmings are controlled either from below or from above – but not from *both* below and above. This is an old-standing issue in ecology: whether populations are regulated from above (Hansson 1987, Hanski et al. 1993, Korpimäki 1994, Korpimäki and Krebs 1996, Korpimäki and Norrdahl 1998) or from below (Hansson 1984, 1987, Batzli 1992, Batzli et al. 1989; see also Hairston et al. 1960, Oksanen et al. 1981). Studies on multiannually fluctuating small rodents will continue to contribute to this part of ecology.

Much information (e.g., Korpimäki 1993, Korpimäki and Krebs 1996, Stenseth et al. 1996a) – by and large not reviewed in this paper – suggests that specialist predator-prey interaction may be a good candidate for explaining the cycles – at least in Fennoscandia. It should be emphasised though that this proposition is in desperate need of experimental testing. Some of my colleagues are advocating the predator-hypothesis in a too dogmatic

³ Reducing Eq. (6) to the delay formulation given by Eq. (7) is easily done by solving the first equation of Eq. (6) for y and substituting this into the lower equation in Eq. (6), and then remembering to get the time indexes right.

⁴ I have some difficulties with both the Oli and Dobson (1999) and the Wikan and Mjølhus (1995) models, but do include them here nevertheless to show that in fact such models have been reported. The problem I have with Oli and Dobson (1999) is that they have incorporated a phase-dependent factor in the demography, the detailed effects of which is not at all clear to me. The problem I have with Wikan and Mjølhus (1995) is that they essentially assume unrealistic life spans to be applicable to small rodent populations.

– and unjustified – fashion, a behaviour which should not be found in scientific communities. In my mind the proponents of the predator-prey hypothesis often fall into the long, but unfortunate, tradition within small rodent ecology of promoting their point of view far beyond theoretical and empirical support – and to block criticism of their view. More work on the mechanistic interpretation of time-series models are needed. However, above all more experimental studies on the individual level (focusing on demography) are needed. Results of such experimental work should, I strongly contend, be interpreted within the framework of models such as, for instance, the one reported by Jorde et al. (unpubl.) [a model which was originally developed for evaluating the dynamic consequences of the maternal effect hypothesis (see, e.g., Boonstra 1994), but which easily may be extended to incorporate other hypotheses]. An important set of experiments have been, and are currently, carried out by Korpimäki and his colleagues (Klemola et al. 1997, 1999, Korpimäki and Norrdahl 1998, Klemola 1999).

Hanski and Henttonen (1996) and Hansen et al. (1999b) extended the study of the density-dependent structure of periodically fluctuating vole populations by dissecting intra- and interspecific competitive effects. Hansen et al. (1999b) showed that observed direct annual density dependence may be split into two approximately equally strong components of intra- and interspecific annual density dependence, whereas no delayed density-dependent effect from other coexisting rodents could be detected. At a more general level, this study suggests that if we are to understand the rodent cycles, we should take a species-specific point of view rather than treating all rodents as a homogeneous group [as is done by some; e.g., Turchin (1993)]. However, the rodent species we focus on should be seen in a community context (including both other competing species, as well as either its food species or its predator specialist – but not both).

CODA: The temporal dynamics of multiannually fluctuating voles and lemmings correspond to an order-two process either due to a specialised predator-prey system or to a specialised food-herbivore system (but not both) – most probably the former even though this remains conjectural. Experimental work testing the many propositions relating to the importance of predation is very much needed, though. Available theoretical results suggest the importance of studying the interaction between the deterministic and the stochastic component of the interacting system – a challenge which certainly will need the involvement of several professions. In fact, the stochastic component is at present hardly discussed at all in the literature on small rodent cycles – it is generally treated as a black box (within which we put all we do not understand nor have data on), and left at that (see below).

Non-linearity caused by phase dependence in the annual density-dependent structure

Pure deterministic and *linear* systems cannot by themselves (i.e., in a hypothesised deterministic world) generate sustained population cycles (e.g., Royama 1992). However, by adding white noise (i.e., some sequence of independent random variables with mean zero and fixed variance), approximate cyclic behaviour with the properties seen in vole and lemming cycles will emerge (see, e.g., Stenseth et al. 1996b): the period will be relatively fixed, whereas the amplitude will vary greatly from one cycle to another (see above). From this we might draw the conclusion that the observed vole and lemming cycles may be a product of the interaction of system-intrinsic deterministic process and system-extrinsic (e.g., environmental) stochastic processes.

Linear – or rather log-linear [see Eq. (4)] – models are necessarily approximations to the real world since the world clearly is non-linear (there is, for instance, a finite amount of resources so that no population can grow infinitely). What is unclear though is the extent to which underlying mechanisms depart from linearity – or log-linearity. The fact that the small rodent density cycle tends to be asymmetric [with a long low and increase phase, and a rapid decline phase (see, e.g., Boonstra et al. 1998)] suggests some underlying non-linearity in the generating process. Investigating questions relating to the extent of non-linearity and where in the interacting system such non-linearity is located, is important. Such studies might indeed help us disentangle properties of the mechanisms being responsible for the generation of the non-linearity in the system.

Assuming a specialised predator-prey model, it can be shown (see Stenseth et al. 1996a, Chan et al. unpubl.) that the first lag [f_1 in Eq. (4)] corresponds to intrinsic interactions within the interacting populations whereas the second lag [f_2 in Eq. (4)] corresponds to both within-population components and external population interactions (such as trophic interactions; see, e.g., Eqs (5)–(7)). Hence, finding non-linearity only in the second lag, it can be argued that the non-linearity resides in the predator-prey interaction as such (for an example, see Stenseth et al. 1998c). However, if the non-linearity is found only in the first lag, we can conclude that the non-linearity may be due to non-linearity in the vole-intrinsic and/or the predator-intrinsic components of the system. Analysing the data on the grey-sided vole from Kilpisjärvi in northern Finland, Chan et al. (unpubl.) have demonstrated that the non-linearity was located in the first lag, suggesting that the observed non-linearity may be due to some sort of social process within either the vole population or the predator population (or both). I should add though, that here we need more work – both empirically based work as well as more methodologically oriented studies.

Dealing with non-linearity is not easy. From the point of view of statistical parsimony, the shortness of the rodent time series precludes complex models. Turchin (1993) adopted a pure curve fitting exercise using a polynomial approximation. Ellner and his colleagues (McCaffrey et al. 1992, Nychka et al. 1992) used an approach based on neural nets [see also Ellner and Turchin (1995)]. Both approaches are difficult to interpret biologically. I am yet to be convinced that the approaches of Turchin and Ellner are very helpful tools for improving our biological understanding of the rodent cycle. Incorporating non-linearity by applying a threshold autoregressive model (TAR model; see, e.g., Tong (1990, 1995)) makes, in my experience, ecological interpretation easier (see, e.g., Framstad et al. 1997, Stenseth et al. 1998c, d, 1999; see also Stenseth and Chan 1998). The TAR approach essentially corresponds to assuming a piecewise linear representation of the underlying non-linear model. Hence, we may for the different TAR regimes (i.e., linear pieces) interpret the statistical coefficients utilising the framework of Eq. (7). The parameters of TAR models may furthermore be estimated using well-developed statistical methods (see Tong 1990). Such models also have a rich spectrum of dynamics (see, e.g., Stenseth et al. 1998c); see also Fig. 5 below.

The TAR model used when analysing the data on the grey-sided vole at Kilpisjärvi (see, e.g., Henttonen and Hanski 1999; see also Fig. 2A) is given as:

$$x_t = \begin{cases} \theta_{1,0} + \theta_{1,1}x_{t-1} + \theta_{1,2}x_{t-2} + \sigma_1\varepsilon_t & \text{if } x_{t-d} \leq r \\ \theta_{2,0} + \theta_{2,1}x_{t-1} + \theta_{2,2}x_{t-2} + \sigma_2\varepsilon_t & \text{otherwise,} \end{cases} \quad (8)$$

where the parameters, $\theta_{i,j}$, are the autoregressive parameters for the two regimes, σ_i are positive parameters, the size of which signify the degree of stochasticity in the two regimes, and ε_t is a noise term

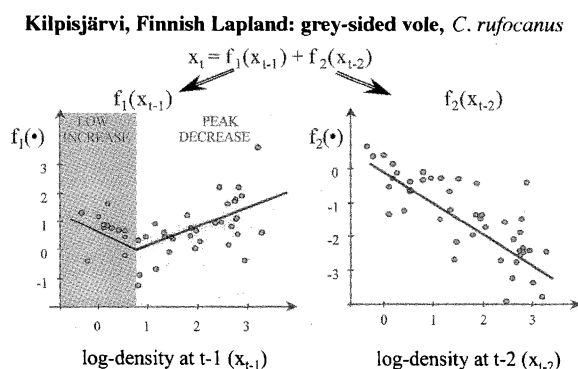


Fig. 4. The non-linear continuous TAR model for the grey-sided vole (*Clethrionomys rufocanus*) in Kilpisjärvi, northern Finland (Chan et al. unpubl., Stenseth et al. unpubl.). Notice that the non-linearity only is located in the first lag. The threshold is argued (Chan et al. unpubl.) to represent the separation between the low/increase phase and the peak/decrease phase.

represented by a normally distributed quantity with mean equal to zero and variance equal to one (i.e., white noise), the effect of which may be different for the different regimes (σ).

The choice of using a TAR model was important since it turned out empirically that the observed non-linearity, and the thresholding, can be interpreted as *phase dependences* in the annual density-dependent structure: that is, during the increase phase (found to correspond to the lower regime) the density-dependent structure is different from that during the decrease phase (found to correspond to the upper regime); see Fig. 4. [Earlier this had been found for lemmings (Framstad et al. 1997, Stenseth et al. 1998c).] This further allows us to interpret the non-linearity in a more biological way – and in fact in a conceptually similar manner to much of the earlier arguments on phase dependences by Chitty (1960, 1967, 1996, Krebs 1978, 1996), but without saying, as they did, that the phase dependence (i.e., non-linearity) is by itself causing the regular density cycle. Their argument was that there existed two or more different kinds of individuals within the population which were optimally adapted to cope with conditions under different phases of the population cycle (a polymorphism which incidentally will lead to phase dependence and non-linearity at the population level). I have earlier demonstrated that such a polymorphism (the Chitty hypothesis) by itself is unable to cause cycles (Stenseth 1981, Warkowska-Dratnal and Stenseth 1985, Stenseth and Lomnicki 1990). The intuitive reason for this is simple: essentially the Chitty hypothesis assumes a competitive system – and competitive systems are generally known to be rather stable (see, e.g., Maynard Smith 1974).

Indeed, the non-linearity and the phase dependence may in themselves lead to sustained multiannual density fluctuations (e.g., May 1974; see also Stenseth and Chan 1998) – but the nature of the non-linearity in small rodents is such that they do not (see also Stenseth 1977a, 1981). That is, the proposition (e.g., the Chitty hypothesis; see Chitty 1960, 1967) long favoured by members of the Chitty–Krebs school cannot be supported by empirical findings (see, e.g., Krebs 1978, Chitty 1996). Experience further suggests that for small rodent cycles to occur, we need some second-order lag – linear or non-linear (e.g., in the form of phase dependence).

CODA: Non-linearity can, for multiannually and periodically fluctuating populations (e.g., voles, lemmings and lynx), properly be interpreted as a phase-dependent structure in the density-dependent structure. Hence, we may refer to phase and annual density-dependent population dynamics, or equivalently a phase-dependent annual density-dependent structure in the observed population fluctuation. It is, however, the delayed annual density-dependent structure – not the phase-dependent structure – which drives the cycle.

The importance of stochasticity in a non-linear world

Most practising field ecologists appreciate the importance of stochasticity (be it sampling error or demographic and/or environmental stochasticity) when working in the field and when interpreting their field data. However, when discussing population models, essentially only deterministic models are considered – a tradition which presumably relates to the fact that Lotka (1925, 1932) and Volterra (1926) worked with deterministic differential equation type of models. Even though some attempts have been made at incorporating stochasticity (particularly demographic and environmental stochasticity) in population models (see, e.g., Bartlett 1970, Skellam 1972, May 1973, 1981), most population models discussed in the literature today are deterministic. [Notice, however, that population genetics developed differently by emphasising stochasticity as one of the key factors in the process of evolution (see, e.g., Maynard Smith 1989).]

Stenseth et al. (1998c) focused on the interaction between deterministic and stochastic processes in a non-linear system. The model was a continuous TAR model using $r_t = x_t - x_{t-1}$ as the basic variable and defined as:

$$r_t = a + \begin{cases} c(r_{t-1} - \psi) + \varepsilon_{(-),t} & \text{if } (r_{t-1} - \psi) \leq 0 \\ -k(r_{t-1} - \psi) + \varepsilon_{(+),t} & \text{if } (r_{t-1} - \psi) > 0, \end{cases} \quad (9)$$

where a , c , k and ψ are parameters (see, e.g., Stenseth et al. 1998c) and $\varepsilon_{(-),t}$ and $\varepsilon_{(+),t}$ are some stochastic variables. The parameter ψ represents the threshold defining the phase, whereas c and $-k$ represent the annual density dependences within the two regimes.

Numerical analysis of the model given by Eq. (9) indicates that the full non-linear model including stochasticity has a somewhat longer period than the underlying deterministic skeleton model (not including the stochastic component, ε) (Fig. 5). Furthermore, in the parameter regions for which the deterministic skeleton shows chaotic dynamics, the full model including stochasticity exhibited a behaviour which cannot easily be distinguished from regions in parameter space for which the skeleton exhibits limit cycle type of dynamics. That is, introducing stochasticity into an otherwise deterministic model (the skeleton) may provide new and important emergent properties. We should of course be careful not to generalise too much from this single study. However, the analysis summarised in Fig. 5 does indeed suggest that it might be quite rewarding to study the *interaction* between the deterministic and the stochastic components of the interactive system. [See Stenseth et al. (1998c) for further details; see also Bjørnstad et al. (1999c).]

CODA: It is unwise to ignore the stochastic component of the system under study since important emergent

properties may arise from the interaction between the deterministic and stochastic processes. Ignoring stochasticity furthermore implies that we – without further analysis – ignore potentially important components of the ecological system, be it within the population or relating to the population-environment interaction.

The geographic patterns in dynamic features

(a) Spatial gradients in the annual density-dependent structure

Since Hansson (1971) first reported on the presence of only seasonal dynamics in southern Sweden for an otherwise multiannually cyclic species (in Fennoscandia), the field vole (*M. agrestis*), it has been extensively documented that in Fennoscandia any given species with a wide geographic distribution tends to exhibit periodical and multiannual density fluctuations superimposed on seasonal density cycles north of 60°N, whereas further south the same species only seem to exhibit seasonal density fluctuations (see Hansson and Henttonen 1985, 1988, Henttonen et al. 1985).

The seasonal density cycles observed in nearly every small rodent species can in most cases be ascribed to the seasonal reproductive cycle. The multiannual density cycles being superimposed upon the seasonal density cycle seen north of 60°N is the core topic of this review. As it appears from the preceding discussion, the causes of the periodic and multiannual fluctuations superimposed upon the seasonal dynamics are unclear, even though specialist trophic interactions seem to be a very good candidate.

Hanski et al. (1991) further documented that within the geographic region of multiannual cycles (i.e., north of about 60°N) there is an increase in the cyclic period as going north from 60°N: around 60°N the cyclic period is about three years, whereas in the northernmost part of Fennoscandia cycles are about five years. Bjørnstad et al. (1995) extended this result by demonstrating that this gradient in periodicity may be seen to result from a decreasing strength of the direct annual density dependence, whereas there was no similar gradient in the delayed annual density dependence (see Fig. 6). It has furthermore been observed (see, e.g., Hanski et al. 1991, 1993) that the amplitude also tends to increase as going north from approximately 60°N; as pointed out by Hansson and Henttonen (1985, 1988) this increase in amplitude as going north is associated neither with higher average densities nor with higher peak densities, but seems to be due to the low densities in the north being lower than further south. The reason for this is unclear.

Stenseth et al. (1996b) reported a somewhat similar geographic gradient in the annual density-dependent

Finse, Southern Norway: Norwegian lemming, *Lemmus lemmus*

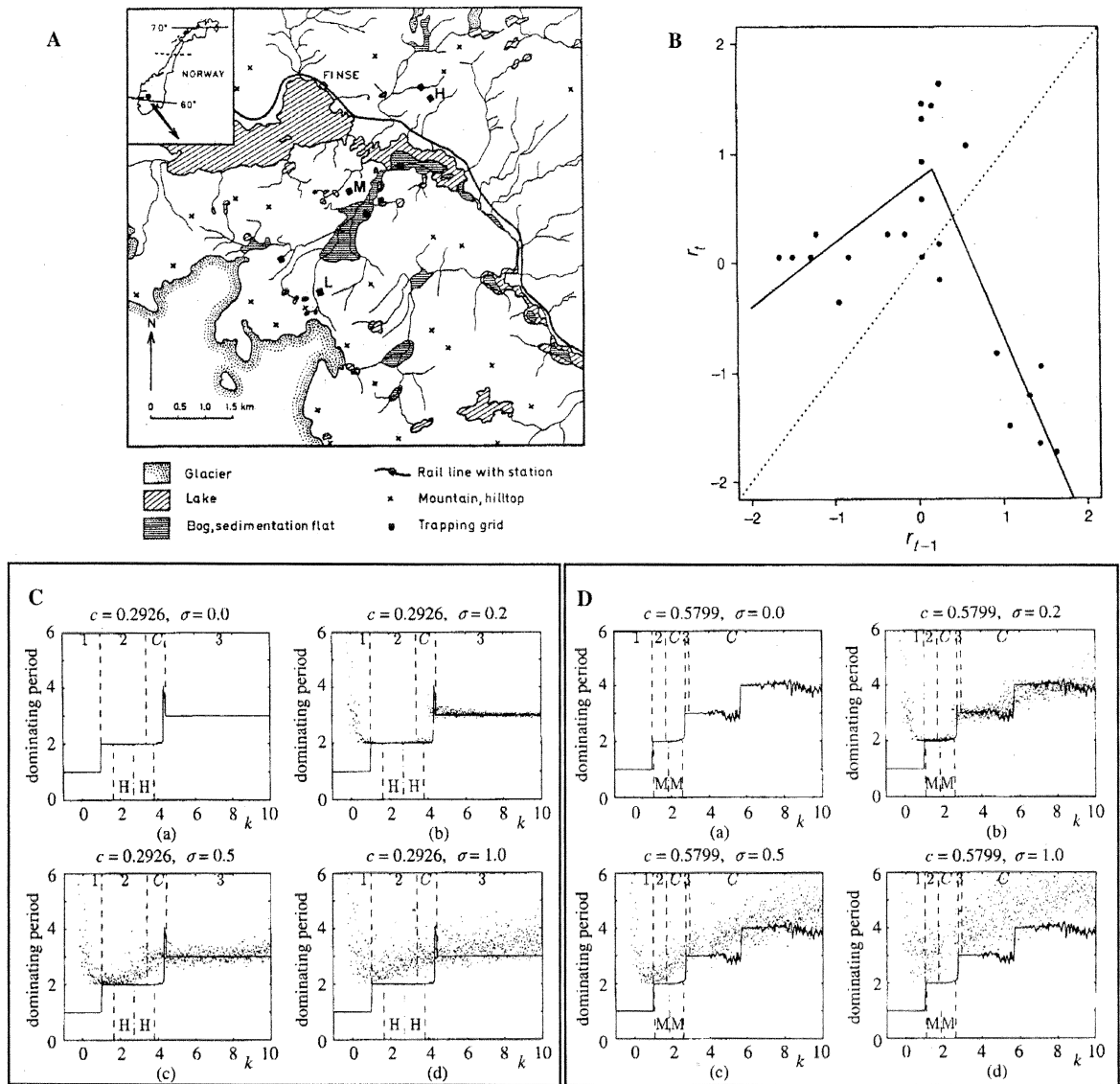


Fig. 5. The non-linear continuous TAR model fitted to the survey data on the Norwegian lemmings (*Lemmus lemmus*) from Finse, Norway; the dynamic variable modelled being $r_t = x_t - x_{t-1}$, where x_t is the log-transformed abundance index (for further details, see Stenseth et al. 1998c). A: the geographic area from which the data were derived; H being the high-productive site and M the medium-productive site [in addition a low-productive grid (L) was studied, but not analysed due to too erratic density changes with too many zeros in years between peak years (for details see, Framstad et al. 1993a, b)]. B: the fitted TAR model. The sets of panels in C and D show the dynamic behaviour of the model for some selected sets of parameters (for a full description, see Stenseth et al. 1998c), both with and without the effect of stochasticity. The parameters c and $-k$ represent the annual density dependences in the two regimes (see the main text): $c = 0.2926$ corresponds to the estimated value for the high-productive grid (H) and $c = 0.5799$ corresponds to the estimated value for the medium-productive grid (M); hence panel C corresponds to the H-grid and panel D corresponds to the M-grid. The 'C' in the upper part of panels C and D indicate chaotic dynamics of the skeleton model (i.e., when not incorporating stochasticity). The 'H' in the (C) diagram indicate the region corresponding to the confidence interval of the estimated value of the parameter k ; the 'M' in the (D) diagram indicate similarly the region of the corresponding confidence interval for the k -estimate. As can be seen, under the influence of stochasticity interacting with the deterministic skeleton is to increase the period over what is expected from the deterministic skeleton model. In addition, regions defined as chaos in the deterministic model become non-indistinguishable from non-chaotic regions under the influence of stochastic processes. [Figure adopted from Stenseth et al. (1998c).]

structure of the grey-sided vole (*C. rufocanus*) in Hokkaido in Japan. In Hokkaido there is a gradient extending from the south-western to the north-eastern part of the island (see also Saitoh et al. 1998): in the south-western part there is only seasonal population dynamics whereas further north-east there is typical multiannual density cycles; in the south-western part both the direct annual density dependence and the delayed annual density dependence are weak, whereas further north-east they are both stronger (Fig. 7).

Even though the overall geographic patterns both in Fennoscandia and in Hokkaido are fairly clear, there are many subtleties. For instance, in Hokkaido there seems to be a region in the middle of the island with particularly long cycles (Bjørnstad et al. 1996, 1998b); this I will return to below when discussing the effect of seasonality and the relative lengths of summers and winters.

A somewhat similar geographic gradient has recently been documented for the muskrat (*Ondatra zibethicus*) (Erb et al. 1999). In the eastern part of Canada, there is a 3–4-year multiannual cycle, whereas further west there is an approximately 8–9-year-long cycle (Errington 1943, 1946a, b, 1961, 1963, Boyce 1977, 1978). This example is interesting since it suggests that similar geographic gradients to those observed in both Fennoscandia and Japan (Hokkaido) also may be found in North America. Much more work remains to be done here, some of which is already in progress (see, e.g., Erb et al. 1999, Yao et al. unpubl.).

The existence of gradients in population dynamic features are interesting in themselves and add to other macroecological patterns (see, e.g., Brown 1995, 1999, Lawton 1996, 1999, Gaston and Blackburn 1999). Explaining the features of these gradients may indeed provide a key to the explanation of the multiannual cycles. The fact that we both have well-studied populations with only seasonal density cycles and populations with both seasonal and multiannual density cycles, also provide a very good platform for experimental work designed to manipulate populations so as to see whether we can change their dynamic characteristics from cyclic to non-cyclic and vice versa. It might, for instance, be worth trying to make a population originally exhibiting both seasonal and multiannual cycles to only exhibit seasonal cycles in density – and similarly to make an only seasonally fluctuating population exhibit both seasonal and multiannual density cycles.

CODA: Regional differences – and gradients – in population dynamic characteristics are now well-documented for several Fennoscandian small rodent species and for the grey-sided vole in Hokkaido. The existence of such geographic gradients should be exploited more fully in experimental work trying to

change the dynamics typical of a region to a pattern characteristic of another region.

(b) Spatial scale

Studying the population biology of bank voles (*C. glareolus*), Steen et al. (1996) documented a larger geographic scale of synchrony in density fluctuations than reasonably can be expected on the basis of population intrinsic spatial processes such as, for instance, dispersal (see also Ims and Steen 1990, Steen et al. 1990, Ranta and Kaitala 1997, Ranta et al. 1997a, c): a spatial scale of 30–40 km was found. Later, Bjørnstad et al. (1999a) documented a similar spatial scale for the grey-sided vole (*C. rufocanus*) in Hokkaido. In a Scottish study on the field vole (*M. agrestis*), MacKinnon (1999) found a much smaller spatial scale (13 km). Why we find this difference between the two studies on the grey-sided vole and the one on the field vole is unclear to me; this issue obviously needs further comparative investigations. In connection with this, it is worth pointing out that it has been claimed that the scale of the synchrony seems to have been much larger before the claimed disappearance of the cyclic pattern in northern Fennoscandia (Henttonen et al. 1977, Andersson and Jonasson 1986, Henttonen and Kaikusalo 1993).

Much remains to be done on the issue of spatial scale in periodically fluctuating small rodents – both empirically and theoretically – but available observations clearly suggest that some external factor seems to play an important role: candidate factors may be nomadic generalist predators (affecting the direct annual density dependence as opposed to specialist predators affecting the delayed annual density dependence) or large-scale climatic factors (Ydenberg 1987, Bjørnstad et al. 1999b; but see Petty et al. 1999). Indeed, the field of spatial scale may be another area within which the study of cyclic small rodents may play another important role. Among the key issues here may be the interaction between non-linear processes (as we do know the small rodent cycles to be) and external stochastic processes (see Fig. 5 and the accompanying discussion). Moran (1953b) made some pioneering contributions on this issue – but only with reference to linear systems. Extending Moran's analysis of linear systems to non-linear systems may indeed be of great general interest, far beyond the study of periodically fluctuating rodent populations (see, e.g., Grenfell and Harwood 1997, Bjørnstad et al. 1999c, Koenig 1999).

Lambin et al. (1998) and Elston et al. (1999) have recently extended the analysis of spatial scale by documenting the existence of travelling waves in field voles (*M. agrestis*) moving at a rate of about 20 km per year. Much remains to be done here; however,

the studies by Lambin et al. (1998) and Petty et al. (1999) expose a whole range of new challenges – both with respect to data as well as with respect to methods of analysis. The spatial pattern may indeed be much more dynamic than we until now have thought.

Personally, I think one of the very best sets of data for approaching the many interesting issues relating to the spatial scale of the population dynamics is the data set on the grey-sided voles (*C. rufocanus*) and the wood mice (*Apodemus* spp.) in Hokkaido (see Kaneko et al. 1998; see also Krebs 1997, May 1998, Batzli 1999, Lidicker 1999). Much remains to be done with respect to the analysis of these data; my guess is that they in 50 years from now (or sooner) may reach the status of a benchmark data set for studying spatio-temporal processes in population ecology just like the Canadian lynx

data from the Hudson Bay Company is today (see, e.g., Bulmer 1974, Royama 1992, Stenseth 1995b, Stenseth et al. 1997, 1998d, 1999, Ranta et al. 1999b; see also Keith 1963, 1990, Krebs et al. 1995).

CODA: Even though the patterns are not all that clear, it seems beyond doubt that the spatial scale relating to synchrony is larger than can be explained by population intrinsic factors such as movement between local populations: some sort of external synchronising process seems necessary. The existence of a good empirical platform for the grey-sided vole on Hokkaido may serve as a gold mine for statistically oriented ecologists who want to study spatio-temporal patterns and thus help us understand the spatio-temporal patterns and dynamics better than we presently do.

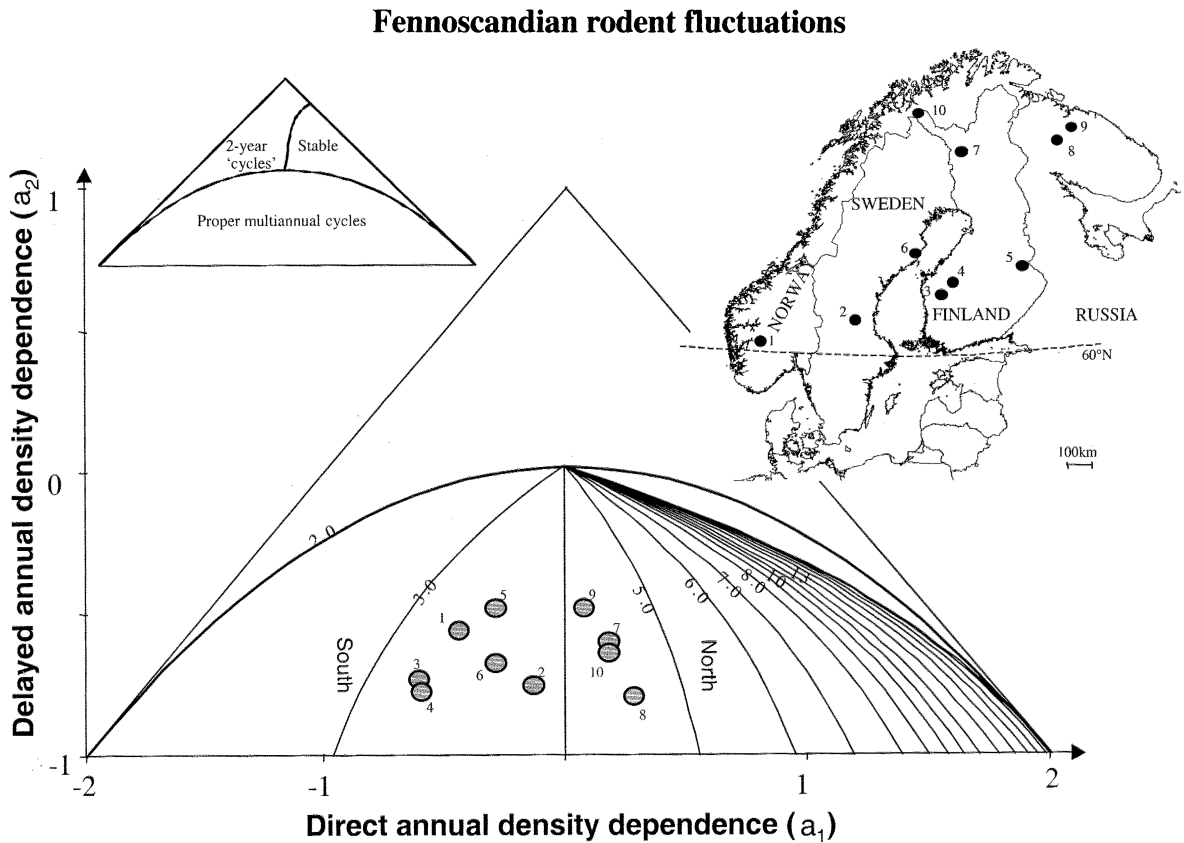


Fig. 6. The location of the estimated first- and second-order coefficients [a_1 and a_2 ; see Eq. (1)] for the Fennoscandian vole and lemming populations [after Bjørnstad et al. (1995) and Stenseth et al. (1996a)]. The broken line on the map indicates the 60°N latitudinal line. The species, locations and sources are as follows: 1) *Lemmus lemmus* and *Microtus oeconomus* at Finse (from Framstad et al. 1993a, b, 1997); 2) *Clethrionomys rufocanus* at Boda (from Marström et al. 1990); 3) and 4) two *Microtus* species at Alajoki and Ruotsala [from Korpimäki and Norrdahl (1991), Korpimäki (1993, 1994)]; 5) *Clethrionomys glareolus* and *Microtus agrestis* at Sotkamo [from Henttonen et al. (1977), A. Kaikusalo (pers. comm.)]; 6) *C. rufocanus*, *C. glareolus* and *M. agrestis* at Umeå [from Hörnfeldt 1978, 1994]; 7) *C. rufocanus*, *C. glareolus* and *C. rutilus* at Pallasjärvi [from Henttonen et al. (1977, 1987), Hanski and Henttonen (1996), H. Henttonen (pers. comm.)]; 8) *C. rufocanus* on Kola (from Kataev et al. 1994); 9) *C. glareolus* on Kola [from Koshkina (1966); data provided by Peter Turchin]; and 10) *C. rufocanus* at Kilpisjärvi [from Kalela (1957), Laine and Henttonen (1983, 1987), H. Henttonen (pers. comm.)]. As demonstrated by Bjørnstad et al. (1995) there is a south-north gradient along the first autoregressive axis (a_1) whereas there is no similar gradient along the second autoregressive axis (a_2); this is indicated by 'south' and 'north' in the triangle. The inserted map shows the locations corresponding to the various species and sites in Fennoscandia. [Figure adopted from Stenseth et al. (1998a).]

Seasonality

(a) Incorporating seasonality in population models

It is a curious fact that even though the typical multiannual vole and lemming cycles seem to be restricted to areas with strong seasonality, characterised by a long winter season during which little biological activity occurs, surprisingly little is done towards incorporating seasonality as such into theoretical models (but see Hanski et al. 1993, Burkey and Stenseth 1994, Turchin and Ostfeld 1997, Hansen et al. 1999a). This is particularly curious since empiricists (e.g., Formozov 1946, Shvarts 1963, 1977, Fuller 1967) have been writing extensively on the importance of the winter (and hence seasonality). However, when discussing seasonality and density cycles, we should not forget that rather dramatic and regular multiannual density fluctuations also occur in more central and southern parts of Europe where seasonality is not as pronounced as in the north (see,

e.g., Frank 1957, Jobsen 1988, Mackin-Rogalska and Nabaglo 1990, Saucy 1994, Lambin et al. 1999, Salamolard et al. 1999).

Essentially, seasonality implies that the population dynamics switches between two modes – the summer (or main reproductive) and the winter (or non-reproductive or secondary reproductive) season. This switching between two modes introduces an inherent oscillator which potentially may be a cause of the multiannual density cycles seen in voles and lemmings of the north: whether in fact the seasonality *is* the cause of the multiannual density cycles remains an open question – a question which often is dogmatically ignored by many [e.g., the proponents of the predation hypothesis: see, e.g., Hanski et al. (1993) who incorporated seasonality in their model but without discussing its dynamic effects].

Incorporating seasonality in population models of the kind described earlier in this review, would be to write:

Hokkaidian rodent fluctuations

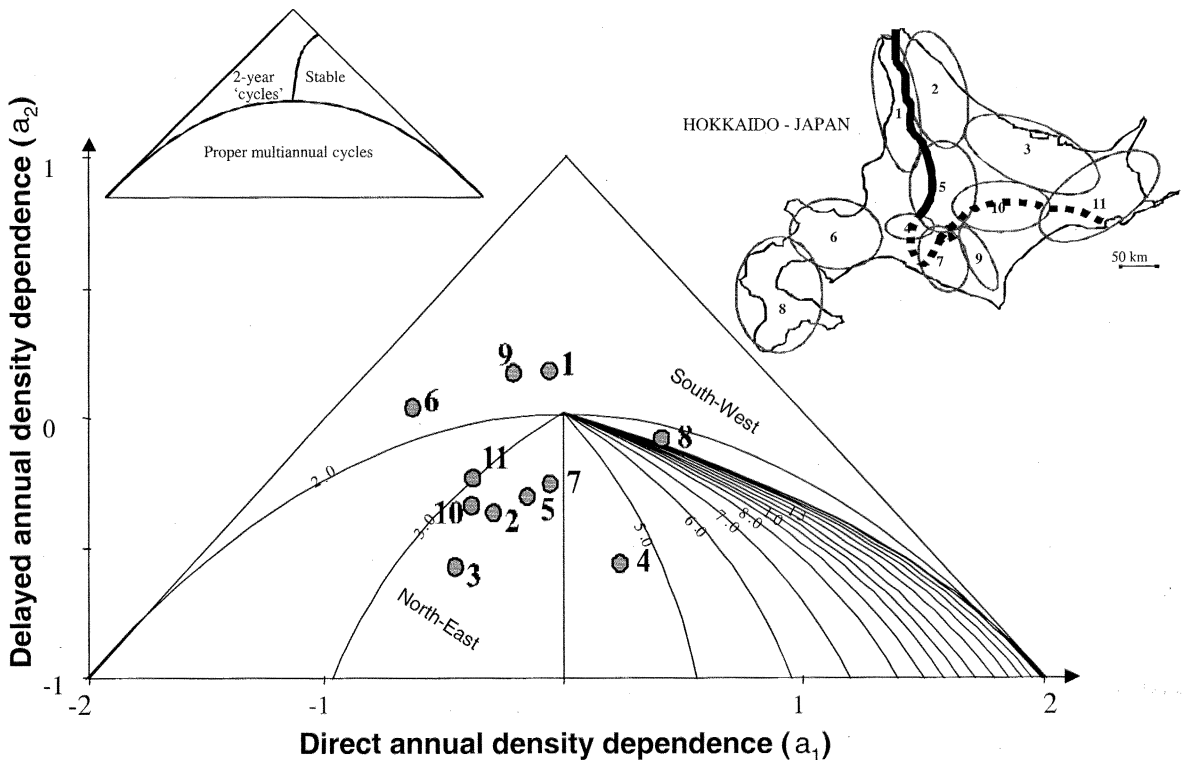


Fig. 7. The location of the estimated first- and second-order coefficients [a_1 and a_2 ; see Eq. (1)] for the Hokkaidian populations of *Clethrionomys rufocanus* [after Saitoh et al. (1998) and Stenseth et al. (1998a)]. The inserted map shows the various regions (see Saitoh et al. 1998, Stenseth et al. 1998a) for which the estimates of the corresponding autoregressive coefficients are plotted in the autoregressive parameter space (the triangular figure); the thick line in the inserted map separates non-cyclic (west) and multiannually cyclic (east) populations of the species in Hokkaido [the broken part of the thick line indicates a more uncertain separation (see Bjørnstad et al. 1998b)]. Analyses are performed on the basis of autumn data, only. The regions are: 1: Teshio for which there are 31 time series; 2: Kitami, 31 time series; 3: Abashiri, 20 time series; 4: Yubari, 8 time series; 5: Kamikawa-Taisetsu, 27 time series; 6: Shiribeshi, 27 time series; 7: West-Hidaka, 20 time series; 8: Oshima, 30 time series; 9: East-Hidaka, 10 time series; 10: Tokachi, 13 time series; 11: Kushiro-Nemuro, 8 time series. The length of the time series are (depending upon the region) between 12 and 31 years (194 are longer than 23 years). [Figure adopted from Stenseth et al. (1998a).]

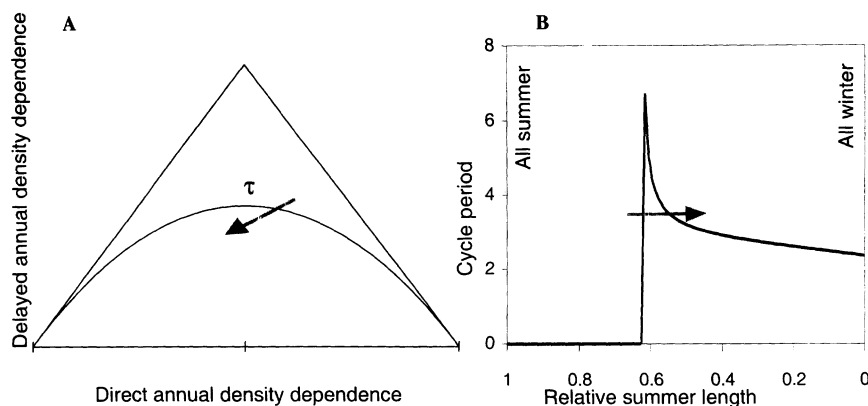


Fig. 8. The effect of length of the summer (τ) where $\tau = 1$ corresponds to all summer and $\tau = 0$ corresponds to all winter. As can be seen from Fig. 7, there is (in the triangular figure) an observed gradient from the upper right part of the triangle towards the lower left part. A similar gradient is produced by decreasing the length of the summer (decreasing τ); see Stenseth et al. (1998a) for details. Notice that the transition goes through a region of expected longer period, just as observed by Bjørnstad et al. (1996). [Figure adopted from Stenseth et al. (1998a).]

$$N_t^{(s)} = N_{t-1} R_w(\cdot) \quad (10)$$

$$N_t = N_t^{(s)} R_s(\cdot)$$

where N_t is (as before) the autumn abundance of year t , $N_t^{(s)}$ is the corresponding abundance in the spring of year t , and $R_w(\cdot)$ and $R_s(\cdot)$ are the net population growth rates during the winter and the summer, respectively. The R functions may have one or more lags depending upon the underlying ecological processes. It is worth noticing, however, that if no reproduction occurs during winter (as is the case for, e.g., the grey-sided vole both in Hokkaido and in northern Finland), $R_w(\cdot)$ will only incorporate survival, whereas $R_s(\cdot)$ will incorporate both reproduction and survival (and possibly also movements). More specifically, we could then write $R_w(\cdot)$ = winter survival, and $R_s(\cdot)$ = [(reproductive rate) (survival of young of the year) + (survival of the overwintered)], the latter of which will (since a negligible fraction of the overwintered survives till the autumn so as to be part of the subsequent overwintering population) appropriately be approximated by $R_s(\cdot)$ = [(reproductive rate) (survival of the young of the year)] (see Stenseth and Saitoh 1998).

Turchin and Ostfeld (1997) modelled the effect of seasonality by the model $N_t = N_{t-1} \exp(b_0(\text{season}) + b_1 x_{t-1})$, where N_t is the density at time t , $x_t = \log(N_t)$, $b_0(\text{season})$ is the maximal seasonal intrinsic net growth rate and b_1 is the direct density dependence. Taking a somewhat different approach – bivariate autoregressive modelling – Hansen et al. (1999a) allowed for all parameters to vary, essentially assuming a model of the form given by Eq. (10). They found that one of the key factors which varied seasonally was the strength of the seasonal direct density dependence: during the non-growing winter sea-

son the seasonal direct density dependence (in the $R_w(\cdot)$ function) was much stronger than during the summer (i.e., in the $R_s(\cdot)$ function) – implying that a major part of the density dependence occurs during the winter. This implies that for the grey-sided vole population at Kilpisjärvi studied by Hansen et al. (1999a), we should expect a particularly strong seasonal density dependence in the winter survival and a relatively weaker seasonal density dependence in the net growth rate [(reproductive rate)(survival of the young of the year)] during the summer. The empirically derived conclusions of Hansen et al. (1999a) thus question the basic premises of Turchin and Ostfeld (1997) who assumed the strength of the density dependence to be independent of season. An important conclusion of the study by Hansen et al. (1999a) is also that some longer lags going two years back need to be incorporated in the $R_s(\cdot)$ and $R_w(\cdot)$ functions in order to describe the seasonal time-series data appropriately, and in a manner which is compatible with the annual time-series structure (see earlier sections).

Assuming that the density-dependent structure varies with the season, Stenseth et al. (1998a) reparameterised the population dynamics model assumed by Stenseth et al. (1996a) and Saitoh et al. (1997, 1998) [$N_t = N_{t-1} \exp(b_0 + b_1 x_{t-1} + b_2 x_{t-2})$ and $x_t = a_0 + a_1 x_{t-1} + a_2 x_{t-2}$; see Eq. (1)] by assuming the per-time specific parameters to be fixed for each season (an assumption which was substantiated through their analysis). Essentially, Stenseth et al. (1998a) assumed that $R_s(\cdot)$ and $R_w(\cdot)$ were of the same form, but with different parameters, and that the relative length of the seasons (τ) could be singled out multiplicatively: i.e.,

$$a_0 = a_{w0}(1 - \tau) + a_{s0}\tau \quad (11)$$

$$a_1 = 1 - a_{w1} + (a_{w1} - a_{s1})\tau, \quad (12)$$

$$a_2 = -a_{w2} + (a_{w2} - a_{s2})\tau, \quad (13)$$

where a_w and a_s are the per time unit (specifically a fraction of the year) density dependences; notice that the a 's are given on an annual basis [see Stenseth et al. (1998a) for further details].

Again the total annual density dependence during the winter was found to be the strongest.

The approach taken by Stenseth et al. (1998a) represented a specification of the approach reported by Åström et al. (1996; see also Kot and Schaffer 1984).

(b) Seasonal forcing

Based on the above annual model structure incorporating the relative length of the seasons (τ), Stenseth et al. (1998a) asked the theoretical question: Assuming a season-dependent fixed density-dependent structure, can we explain the observed geographic gradient in the annual density-dependent structure in Hokkaido by only assuming that the relative length of the summer (and winter) varies between the different sites in Hokkaido [see Fig. 8 and Stenseth et al. (1996a) and Saitoh et al. (1997, 1998)]? This question was answered by a clear *yes* (see Stenseth et al. 1998a). Essentially, they suggested that the cycles may phenomenologically be seen as a result of seasonal forcing (see Fig. 8): if the summer is sufficiently long, there will be no multiannual density cycles – such multiannual density cycles will only arise if the winter is sufficiently long (Fig. 8). It remains unclear though why the density dependence changes between the seasons. In order to answer this we need to develop some mechanism which can explain the seasonal differences in the density-dependent structure. Even though I cannot substantiate it, my expectation is that such a mechanism may include the limiting supply of bamboo shoots (*Sasa* spp.), a major part of the grey-sided voles' winter diet in Hokkaido (Kaneko et al. 1998, Stenseth et al. 1998a), as a critical component. Certainly a lot needs to be done – but analysis till now suggests that such further work may be rather exciting and indeed very rewarding.

As part of the analysis reported by Stenseth et al. (1998a), it was also shown that the relative length of the season (τ) may be a good predictor of the cyclic period length. In fact, the reparameterised model involving the relative length of the season (τ), suggests that after the bifurcation into the cyclic region of the parameter space, there will be longer cycles than where the relative length of the winter increases fur-

ther (see Fig. 8). Interestingly, this was exactly what was observed by Bjørnstad et al. (1996): in the western part of the island there is no multiannual density cycles, in the mountainous region further east there are long cycles (about five-year period), whereas even further east there is a clear three-year cyclic pattern – just as predicted by the model of Stenseth et al. (1998a) (see Fig. 8). It is worth noticing that when we first obtained the pattern reported by Bjørnstad et al. (1996) I – at least – did not fully trust the deduced longer period in the intermediate mountainous region; however, when having a model which predicted this pattern, the statistically deduced pattern seems more trustworthy.

Stenseth et al. (1998a) also showed that the relative length of the season along the north-south axis of Fennoscandia may be a good predictor of the length of the density cycle in Fennoscandia [with short three-year cycles around 60°N and five-year cycles in the far north (see also Fig. 6)].

CODA: We now know that the relative length of the summer and winter seasons may play a key role in the generation of the multiannual density cycle seen in Hokkaido. Indeed, the relative fraction of exposure to winter and summer conditions may keep the key to the enigma of the rodent cycle. The relative length of the season also seems to explain the length of the multiannual density cycle where it occurs in both Hokkaido and in Fennoscandia. Hence, we should focus more on studying the winter ecology of the multiannual fluctuations in voles and lemmings – in addition to studying the rodents' ecology during the summer period: it does not suffice to only study these rodents during the summer period (as is typically done). Again, utilising the opportunities provided by the existence of the gradient in the annual density-dependent structure in both Fennoscandia and in Hokkaido, may prove very helpful. However, it would be narrow-minded to only focus on these gradients. First, further work may expose many exceptions to these overall geographic patterns. Second, other areas (such as, for instance, Central Europe) may provide useful insights over and beyond what we can obtain from the Fennoscandian and Hokkaidian gradients.

A view on the vole and lemming density cycle

On the basis of the studies I have reviewed above, we can conclude:

- The vole and lemming cycles typically exhibit both direct and delayed annual density dependence. The direct annual density dependence may result from a variety of processes, both intrinsic and extrinsic to the population – Bjørnstad et al. (1995) and

Stenseth et al. (1996b) discuss possible mechanisms. (The literature – including many papers in *Oikos* – on the vole and lemming cycles abound in suggested possible factors.) Much evidence suggests that specialised trophic processes are among the more likely candidates. I am myself inclined to accept the idea of specialist predators causing the multiannual density cycle until experimentally falsified (see, e.g., Stenseth et al. 1996a); however, my problem with many of the recent papers on the predation hypothesis (e.g., Hanski et al. 1993, Turchin and Hanski 1998) is that the links between the data and the model assumptions (and parameters) are quite tenuous. It is potentially a viable hypothesis – but it has by no means been demonstrated experimentally, and it should not be implied as an essentially proved hypothesis as some of its proponents seem to do. The annual density-dependent structure is, however, clear: both direct annual density dependence (DD) and delayed annual density dependence (DDD) are always involved (remember that ‘no direct annual density dependence’ corresponds to $a_1 = 1$).

- Interpretations of the non-linear structure of time-series data on voles and lemmings – as well as other cyclic terrestrial vertebrates such as lynx (see, e.g., Stenseth et al. 1998d) – suggest that the non-linearity in the annual density-dependent structure may be expressed as a phase dependence. That is, the processes seems to differ qualitatively between different phases of the cycle. This is essentially what Chitty (1960, 1967, 1996) and Krebs (1978, 1996) have been saying all along, but not expressed in the language of dynamic systems. (This is not to say that the phase dependence drives the cycles – in fact, this is likely not to be the case.) I personally think that interpreting the observed non-linearity as phase dependence in the annual density-dependent structure helps us understand mechanistically what non-linearities really are. In short we may say that the annual DD-DDD structure is phase dependent (PD).
- On the basis of what we know today, seasonality should not be ignored. Nevertheless this is commonly the case in most theoretical considerations of the vole and lemming density cycles; even though seasonality is incorporated in the models, seasonality as such is only rarely focused upon as a key element of the system. In short we may say that the PD-dependent DD-DDD structure is season dependent (SD).
- The interaction between deterministic and stochastic processes is more important than commonly is appreciated in the literature on the small rodent multiannual density cycles. Much work remains to be done, but I am convinced that by taking the interaction between deterministic and stochastic

processes more seriously, we will obtain a better understanding of the vole and lemming density cycle, as well as the underlying population dynamic processes. In short we may say that the PD- and SD-dependent DD-DDD structure should be seen in the light of interactions between deterministic and stochastic processes (D-S-I).

I am quite convinced that the conceptual key to the enigma of the rodent cycle is to be found within the ‘PD-SD DD-DDD D-S-I’ framework – a rather complicated one, but one which is easily taken apart and put together again when we understand its components reasonably well.

I should add a comment on the necessity of merging various approaches to the study of ecological patterns and processes in the vole and lemming cycle. Today we are ready for merging the statistical and the mathematical approach: Statisticians arrive at a description of the structure of the temporal variation [an example is given by Eq. (8)]. Such statistical descriptions will always be an approximation to reality – but they are, if appropriate methods and relatively good data are employed, the best possible approximations we can provide. Typically, statisticians are not too concerned about the emerging dynamics of the obtained model. Mathematicians, on the other hand, assume some model, and on that basis study its dynamical behaviour – but then only with superficial reference to the real world. The challenge is to merge these approaches by linking the statisticians’ – and ecologists’ – best ecological models with the ‘mathematicians’ *assumed* (or hypothesised) ecological model (see, e.g., Kendall et al. 1999). Specifically, this has to be done by calculating the values of the parameters in the mathematical model being compatible with the estimated parameters of the statistical model. And above all, we need to let the data determine what the most appropriate model is to be – and not let our prejudices determine what is the ‘right’ model.

Certainly much more work is needed on this very issue of linking different approaches. We have, however, demonstrated that it is possible to link these traditionally separated approaches. By pursuing this companionship further, we will – I am sure – learn many new things about ecological dynamics, and nature in general. Indeed, this might represent *the* methodological key to the chest holding the solution to the ‘rodent cycle enigma’.

The challenges: the demographic machinery and the population cycle

Essentially all of this review has focused on the an-

nual (and to some extent the seasonal) density dynamics of the rodents and lemmings. It is a non-avoidable fact that the annual and seasonal net growth rates result from demographic processes such as reproduction, survival and movement referring to a much shorter time-scale than the seasonal and annual scale – this because voles and lemmings have several generations within a season (and hence within a year). The challenge ahead of us is to link observed density- and phase-dependent structure in the seasonal and annual density changes to density- and phase-dependent structures in the seasonal demographic rates (survival, reproduction and movement). Essentially, we need to link demography and population dynamics. The Leslie–Usher model represents a platform for such an integration (see, e.g., Leslie 1945, 1948, Caswell 1989, Yoccoz et al. 1998; see above).

Until now this has been a non-approachable field partly due to lack of sufficient large capture-mark-recapture (CMR) data sets. Analyses of such data do, however, suggest that we see a similar signature in the phase- and seasonal-dependent density-dependent structure of the demographic data as we see in the population data (Boonstra et al. unpubl., Stenseth et al. unpubl.). More detailed analysis of this kind is currently underway on the long term CMR data from Pallasjärvi (northern Finland; cf. Henttonen 1997; see also Prévot-Julliard et al. 1999) and other areas.

The Europeans have to some extent been focusing on long-term monitoring of population dynamics (see, e.g., Figs 2, 6). The North Americans have been emphasising the collection of CMR data – but have a shortage of long-term population data. Fortunately, long-term CMR data have been obtained in northern Finland in a region for which the multiannual population dynamic structure is fairly well described. I am convinced that integrating the insight from the analyses of CMR data and population monitoring data into a demographically based Leslie–Usher type of model will yield very interesting results – results which indeed may provide the breakthrough in the study of the rodent density cycle.

CODA: The greatest challenge for the years to come relates to tying together the demographic approach (being the dominating one in North America) and the population approach (being the dominating one in Northern Europe). Even though the outcome of such analyses is impossible to predict, it is quite likely that interesting results will emerge – results which will substantially advance our understanding of the multi-annually periodic fluctuating vole and lemming populations of the north.

Looking to the tropics for new and additional model systems

Lemmings and voles of the northern temperate, sub-arctic and arctic zones have served as model systems for population ecologists ever since Elton wrote his 1924 paper. Over the last few years I have come to be involved in some most interesting tropical rodent systems which may become another set of model systems which indeed may help those of us studying voles and lemming of the north. These tropical systems include the multimammate rat (*Mastomys natalensis*) of Africa (see, e.g., Leirs et al. 1997, Julliard et al. 1999; see also Fig. 9).

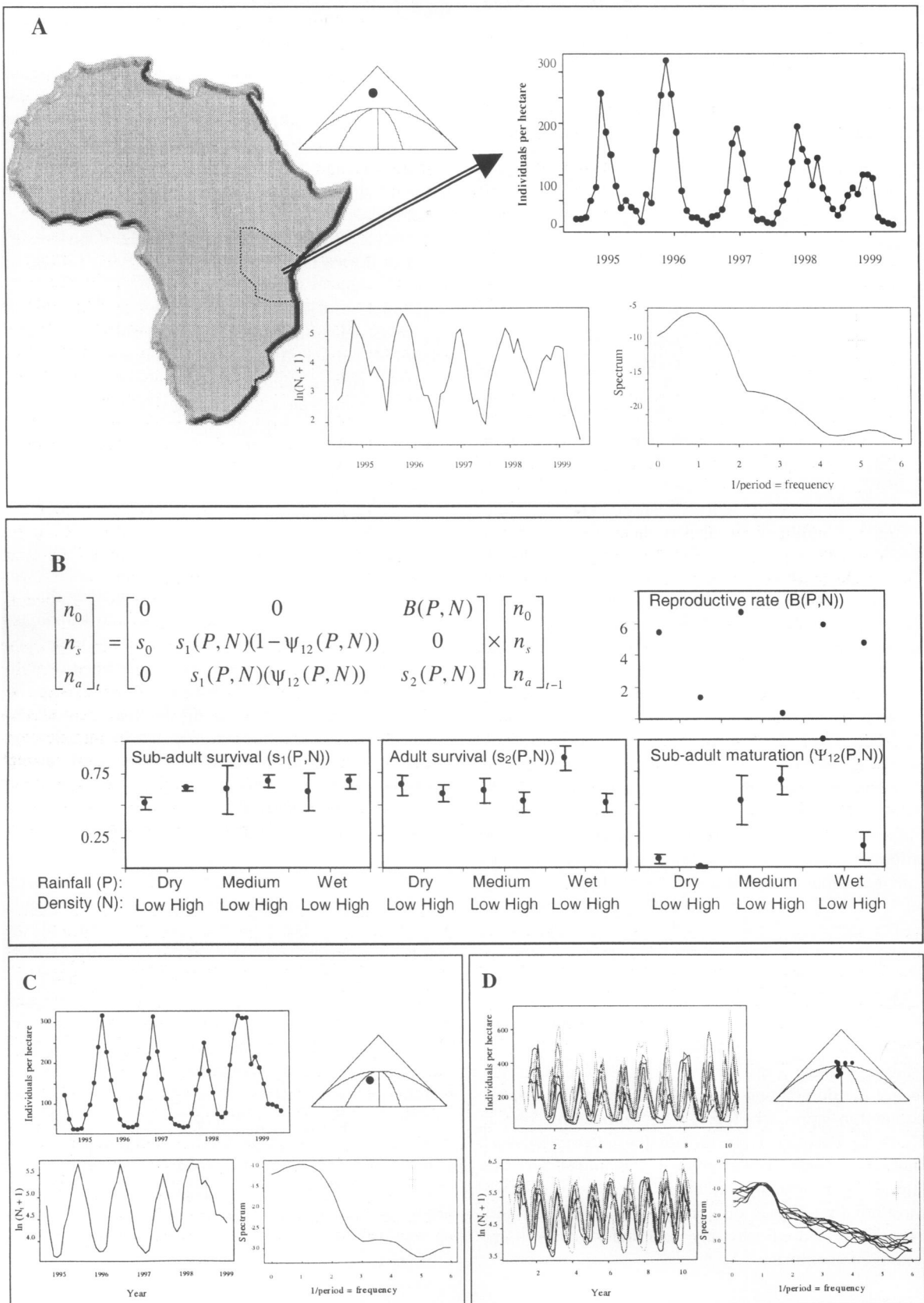
Through CMR-statistical models we have developed a Leslie–Usher type of population model which mimics the observed seasonal and annual population dynamics pattern quite well (Fig. 9). Having such a model we may then ask what the population dynamic effects are of changing the phase- and density-dependent structure of the various elements in the Leslie–Usher model (i.e., the various components of the demographic machinery) for the emerging population dynamic structure. This has – rather successfully (see Fig. 9) – been done for the African multimammate rat. The trouble with this tropical system is that long-term monitoring data do not exist – neither does it exhibit cycles of the kind seen in voles and lemmings.

For lemmings and voles of the north we have such monitoring data – and to some extent long-term CMR data. With our *Mastomys* model as a platform, we are now in the process of developing a CMR-based Leslie–Usher type of population model (e.g., Prévot-Julliard et al. 1999). The emerging population dynamics of such a model may then be compared with the phase and density dependence deduced from time-series modelling (see, e.g., Yoccoz and Stenseth 1999).

With such an integrative approach based on merging demographic and population dynamics, we would also get rid of the (constructed) dichotomy of the biological approach (= demographically based CMR-type of approach) and the numerical approach (= population dynamic studies, including time-series analysis) described by Krebs (1996) and others (see above). Such an integrative approach would furthermore provide an ideal theoretical platform for experimental ecologists interested in working on voles and lemming cycles – not least because the expected population dynamic consequences of manipulating demographic rates thus could be predicted and tested.

This uncovers a methodological challenge – the solution of which will be of great general interest – namely the formulation of CMR models within a time-series context.

Fluctuations in an African rodent, *Mastomys natalensis*



Voles, mice and lemmings will continue to influence ecology another 50 years – at least

One should – in science – be careful about making predictions on where it might be advisable to go next. Breakthroughs too often come from rather unexpected directions. Nevertheless, it seems to me that two approaches clearly are among the most promising ones:

1. Integrate demographic analysis better into the modelling of population dynamics models. Until now the analysis of the vole and lemming density cycle has been carried out in more or less complete isolation from studies on the demographic rates. These two approaches should now be ready for being merged and integrated – for which some sort of Leslie–Usher type of model (see above) will be a natural melting pot.
2. Perform experimental manipulations for populations known to exhibit multiannual cycles (trying to make them only seasonally cyclic) and for populations known to exhibit only seasonal density cycles (trying to make them exhibit multiannual density cycles). Such experiments should aim at identifying factors responsible for the generation of the multiannual density fluctuations superimposed on top of seasonal fluctuations in density. We are at present in a good position for designing such experiments partly due to us having a reasonably good grasp of the patterns which are to be explained, as well as recently having made much progress in the study of the demographic machinery of voles and lemmings. The rodent cycles are such a key issue in ecology with profound applied aspects (e.g., pest control) that it would be worth the effort of population ecologists to join forces, for instance, on the grey-sided vole system of Hokkaido. I am also sure that funding agencies would be wise to support such a consorted effort:

it might be an expensive exercise, but the odds for success would be rather good – provided that a large (multidisciplinary) spectrum of ecologists is ready to collaborate.

I cannot avoid emphasising the need for ecologists to take advantage of the many new mathematical and statistical techniques currently being developed. Furthermore, we ecologists need to be better at making statisticians and other method-oriented scientists get ‘turned onto’ our pet problems. They will, I know, benefit from getting involved, and we – ecologists – will for sure benefit. I put this forward as a challenge, not least due to the difficulties of working in multidisciplinary teams aiming at being in the front on both the method side and on the ecology side. If we are to attract the best people from both sides, the collaborative research group needs to be at the front in both fields. I have only one advice – allow for good time so that each of the participants can get to know each other’s field to a sufficient degree so as to ask more meaningful questions (relating to each of their respective fields) than they otherwise would have been able to ask.

We should in this context remember that interest in the density cycles of the north – and in particular the lynx cycle – started a long and profound development within the field of statistical time-series modelling (cf. Moran 1953a, b). It is time for statisticians, mathematicians and ecologists working on the vole and lemming density cycles (as well as the lynx cycle) to fertilise each other again (cf. Ranta et al. 1999b): statisticians and mathematicians will be exposed to (what to them are) new problems and new sets of data which for sure will continue the further development of methods within these fields – and through such new methods ecologists will come in a better position for interpreting observed ecological patterns. Through such a multidisciplinary joining of forces we

Fig. 9. The population dynamics model for the African multimammate rat (*Mastomys natalensis*) as originally reported by Leirs et al. (1997). A: the observed monthly population dynamics changes [CMR-based density estimates (per hectare) for the study site (Morogoro, 06°51'S 37°38'E) in Tanzania, Africa; June 1994 to April 1999]. Also the log-transformed monthly densities and the periodogram (based on log-transformed monthly densities; compare with Fig. 3 – but notice the scale of the frequency axis) are shown. The inserted triangle is the parameter space for the second-order autoregressive model (as shown in Fig. 1), using values for November, approximately corresponding to the yearly maximal densities. B: The Leslie–Usher type of population model developed on the basis of CMR-based survival (s_i ; $i = 0$ represents juveniles and s_0 is set to 0.5; $i = 1$ represents subadults; and $i = 2$ represents adults) and maturation rates (ψ_{12}); see Leirs et al. (1997). There are no SD for $B(P, N)$, the monthly reproductive rate per adult female, since $B(P, N)$ is the product of litter size, proportion of adult females that are pregnant, corrected for the fact that early pregnancies are not detected in live-trapping and correcting for the fact that trapping happened every four weeks which means that sometimes complete gestation periods (3 weeks) go unnoticed. C: The population dynamics properties emerging from the population dynamics model shown in panel B when the model was run using actual rainfall (P) for the period December 1994 through April 1999; starting population estimates for December 1994 were used as initial values. Both the emerging densities and the log-transformed densities are shown together with the resulting periodogram and the autoregressive parameter estimates using values for September, approximately corresponding to the yearly maximal densities. D: The population dynamics properties emerging from the population dynamics model shown in panel B when the model was run using 10 different bootstrapped rainfall series (see Leirs et al. 1997), and fictive initial conditions (133 subadult females and no adults); the 10 population curves (non-transformed and log-transformed) and the 10 periodograms are superimposed upon each other.

may be able to construct the tools which may help solve the rodent cycle enigma – and by so doing make a profound advancement within population ecology.

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